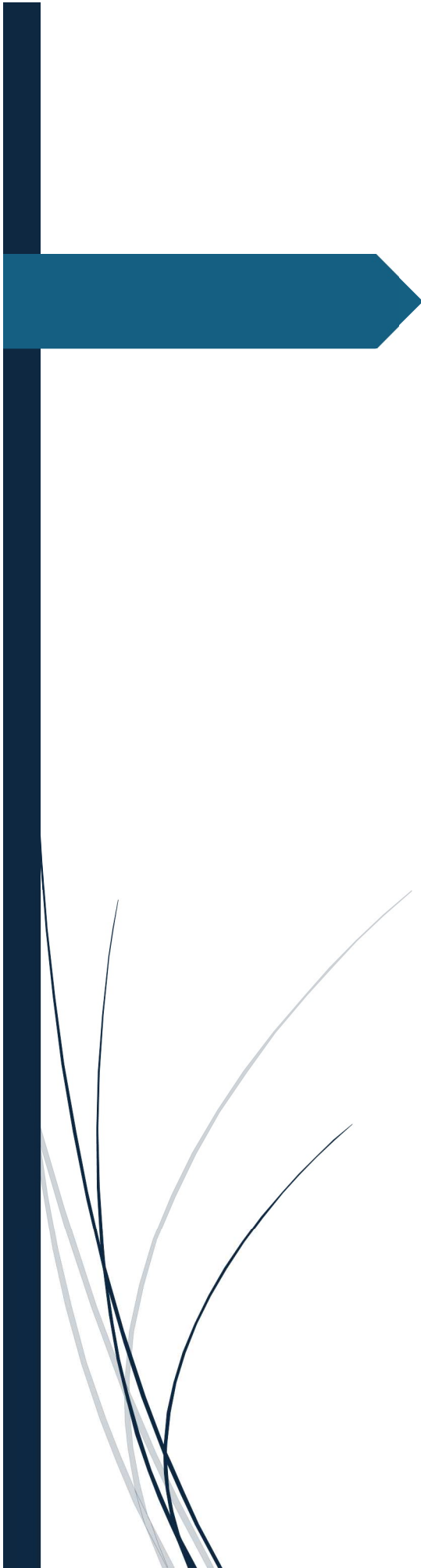




PRINCIPLES OF ECONOMICS

Economics mid-term Notes

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AIUB, Economics Notes

PRINCIPLES OF ECONOMICS

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INTRODUCTION

TOPIC:1

Economics

⇒ Economics is the study of how society manages its scarce resources.

▪ Economics is divided into two subfields:

1. Microeconomics
2. Macroeconomics

Microeconomics

⇒ Microeconomics focuses on the individual parts of the economy. It deals with the behavior of individual economic units.

- Microeconomics explains how and why these units make economic decisions.
- These economic units include consumers, workers, investors, owners of land, business firms.
- How consumers make purchasing decisions, how firms decide how many workers to hire.

Macroeconomics

⇒ Macroeconomics looks at the economy as a whole.

Economy-wide phenomena, including inflation, unemployment, and economic growth.

Economy: In Microeconomics

⇒ The word economy comes from a Greek word “Oikonomia” which basically means “one who manages a household.”

A household and an economy face many similar decisions:

- Who will work?
- What goods and how many of them should be produced?
- What resources should be used in production?
- At what price should the goods be sold?

Economic analysis

⇒ Economists use two types of statements for analysis:

1. **Positive statements**
2. **Normative statements**

Positive statements

⇒ **Positive statements** are statements that attempt to describe the world as it is. Also known as **descriptive analysis**.

Normative statements

⇒ **Normative statements** are statements about how the world should be. Also known as **prescriptive analysis**.

Example of Positive & Normative Statements?

- An increase in the minimum wage will cause a decrease in employment among the least skilled. (**POSITIVE**)
- Higher federal budget deficits will cause interest rates to increase. (**POSITIVE**)
- The income gains from a higher minimum wage are worth more than any slight reductions in employment. (**NORMATIVE**).
- State governments should be allowed to collect from tobacco companies the costs of treating smoking-related illnesses among the poor. (**NORMATIVE**).

Basic Problem of Economics

⇒ There are **three** basic problems:

1. What commodities(product) should be produced and in what quantities?
2. How shall goods be produced (by whom and with what resources and in what technical manners to be produced)?
3. For whom shall goods be produced?

Economic Models

⇒ The most basic economic models include: **The Production Possibilities Frontier**. It is used to solve the basic problems in economics.

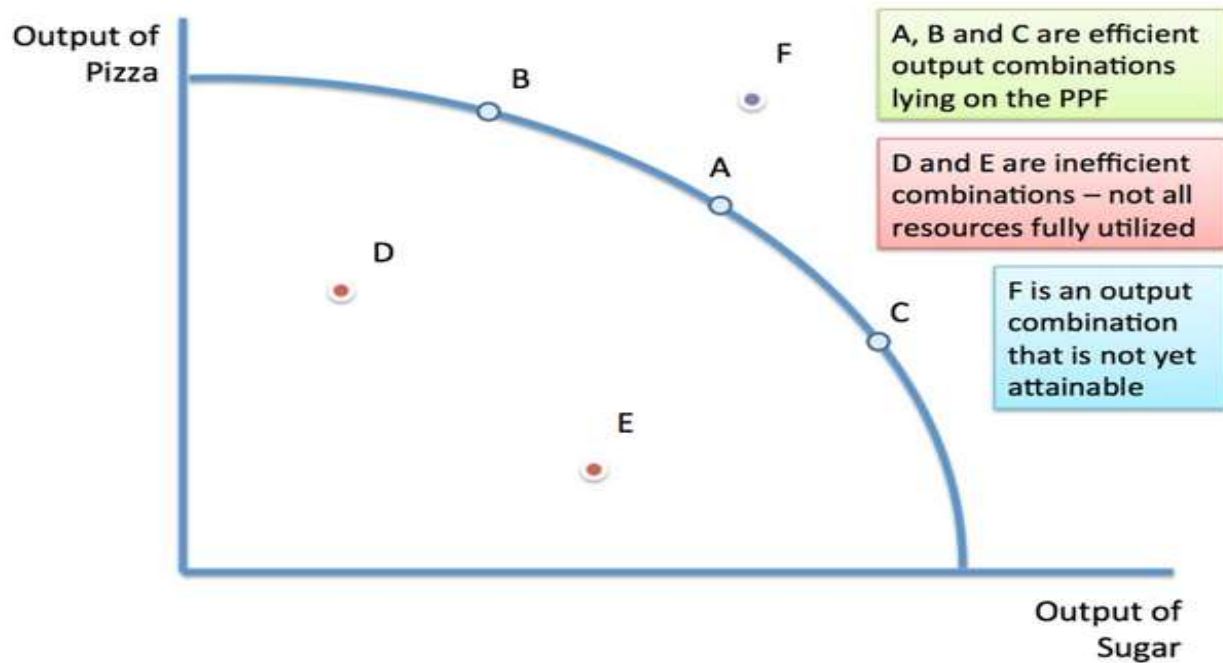
Production Possibilities Frontier (PPF)

⇒ It is a graphical representation of the different combinations of two goods that can be produced with the available resources.

■ Assumption for PPF

1. The amount of resources in an economy is fixed, but these resources can be transferred from one use to another.
2. With the help of given resources, only two goods can be produced.
3. The resources are fully efficiently utilized.

■ Characteristics of PPF:



NOTE: Every point along the curve is efficient, points outside the curve are unobtainable or inefficient. PPF Curve is concave shape.

■ Change in PPF:

⇒ The change in PPF indicates either an increase or decrease in the productive capacity of the economy.

There are 2 types:

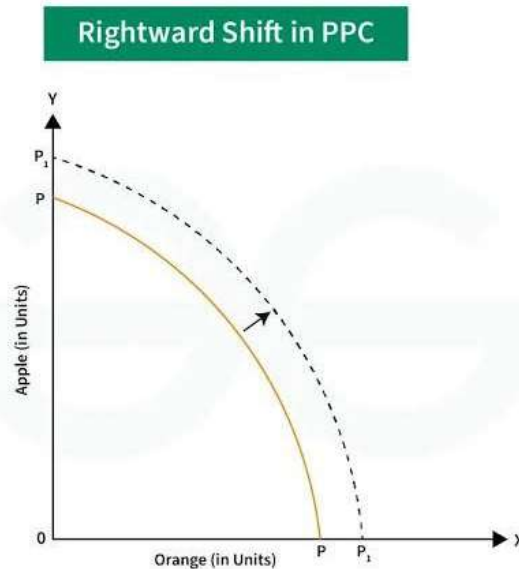
1. Shift in PPF
2. Rotation of PPF

➤ Shift in PPF:

⇒ PPF will shift when there is change in productive capacity (resources or technology) with respect to both the goods.

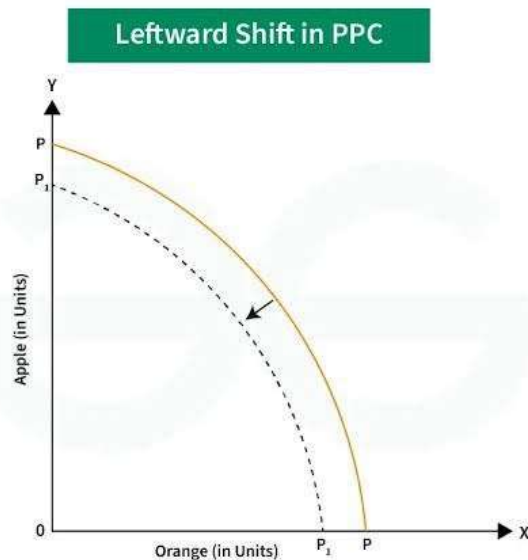
▪ **Right Ward shift in PPF (Outward shift):**

⇒ When there is "Advancement or Upgradation of Technology" or/and "Growth of resources" in respect to both the goods.



▪ **Left Ward shift in PPF (Inward shift):**

⇒ PPF will shift towards left, when there is a technological degradation and/or decrease in resources with respect to both the goods.

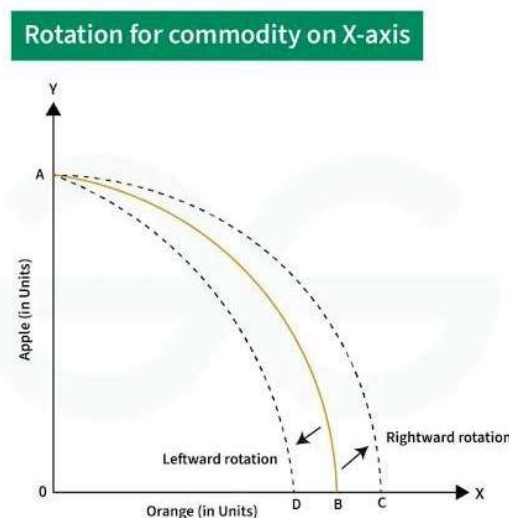


➤ Rotation of PPF:

⇒ PPF will rotate when there is change in productive capacity (resources or technology) with respect to only one good.

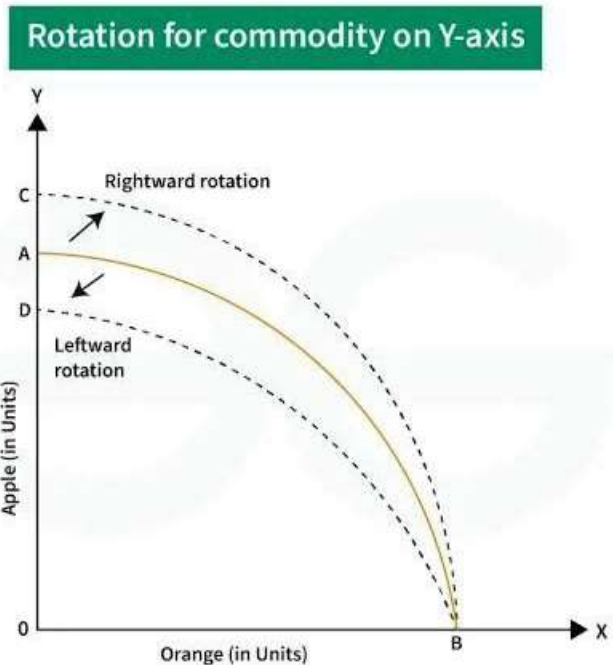
▪ Rotation for commodity on the X-axis:

- ⇒ When there is a technological improvement or an increase in resources for production of the commodity on the X-axis. Its known as RightWard rotation.
- ⇒ When there is a technological degradation or a decrease in resources for production of the commodity on the X-axis. Its known as LeftWard rotation.



▪ Rotation for commodity on the Y-axis:

- ⇒ When there is a technological improvement or an increase in resources for production of the commodity on the Y-axis. Its known as RightWard rotation.
- ⇒ When there is a technological degradation or a decrease in resources for production of the commodity on the Y-axis. Its known as LeftWard rotation.



#Problem 1:

The possible combination of out of an economy of Apple and Orange are given below.

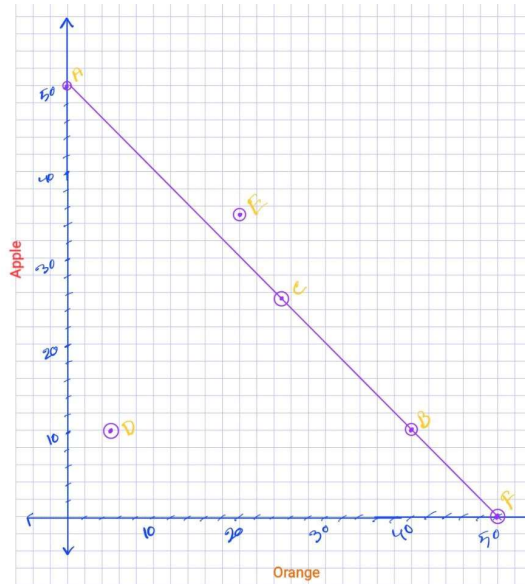
Apple	Orange
50	0
10	40
25	25
10	5
35	20
0	50

- a. Draw the production possibilities frontier using the table.
- b. Analyze which combinations are effective, ineffective, and unattainable. Write down that.

(a)

Combinations	Apple	Orange
A	50	0
B	10	40
C	25	25
D	10	5
E	35	20
F	0	50

Now the PPF curve is:



(b)

- ⇒ The combination of A, B, C, F are effective since they lie on the PPF curve. That means resources are fully and efficiently utilized.
- ⇒ The combination of only D is ineffective since it does not lie on the PPF curve but under the PPF curve. It indicates inefficient use of resources (e.g., underutilization of labor, technology, or capital).
- ⇒ The combination of only E is unattainable because it does not lie on the PPF curve but above it. We do not have enough technology or resources to produce that combination.

#Problem 2:

- a) Draw and Define what is Production Possibilities Frontier (PPF) for an economy that produces mango juice and apple juice.
- b) Indicate and describe an efficient, inefficient, and unattainable point in the PPF with diagram.
- c) What happens to this frontier if the mango trees got destroyed in a storm and some apple trees are newly planted. Explain with diagram

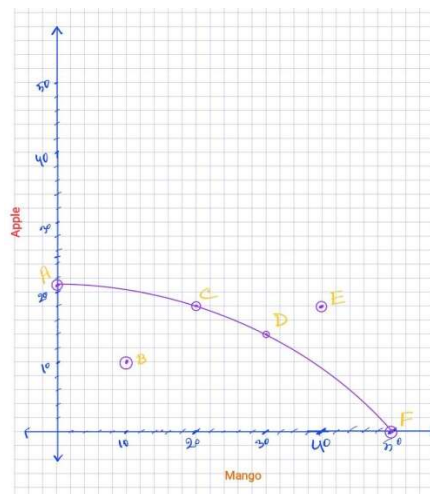
(a)

⇒ PPF is a graphical representation of the different combinations of two goods that can be produced with the available resources.

Let, The hypothetical values for the PPF curve:

Different Combinations	Mango	Apple
A	0	21
B	10	10
C	20	18
D	30	14
E	40	18
F	50	0

Now the PPF curve is:



(b)

⇒ The combination of A, C, D F are effective since they lie on the PPF curve.
That means resources are fully and efficiently utilized.

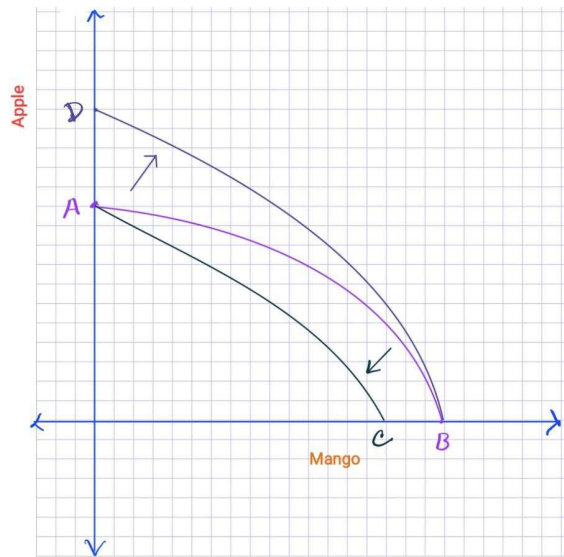
⇒ The combination of only B is ineffective since it does not lie on the PPF curve but under the PPF curve. It indicates inefficient use of resources (e.g., underutilization of labor, technology, or capital).

⇒ The combination of only E is unattainable because it does not lie on the PPF curve but above it. We do not have enough technology or resources to produce that combination.

(c)

Destruction of Mango Trees: The productive capacity for mango juice will decrease, shifting the PPF **inward** for mango juice.

Planting New Apple Trees: The productive capacity for apple juice will increase, shifting the PPF **outward** for apple juice.



- The storm reduces resources (mango trees), causing a decrease in maximum production for mango juice.
- New apple trees increase potential output, expanding the apple juice side of the PPF.
- The new PPF reflects these changes, showing altered trade-offs between the two goods.

SUPPLY & DEMAND

TOPIC:2

THE MARKET FORCES OF SUPPLY AND DEMAND

- ⇒ Supply and Demand are the forces that make market economies work!
- ⇒ Modern microeconomics is about **supply**, **demand**, and **market equilibrium**.

MARKETS AND COMPETITION

- ⇒ The terms supply and demand refer to the behaviors of people as they interact with one another in markets.
- ⇒ A market is a group of buyers and sellers of a particular good or service.

- Buyers determine demand.
- Sellers determine supply.

DEMAND

- ⇒ Demand is the quantity of a commodity that a consumer is **willing** and **able to buy**, at each possible price during a given period of time.

There are Two types:

1. Individual Demand.
2. Market Demand.

Individual Demand

- ⇒ Refer to the quantity of a commodity that **a consumer** is willing and able to buy, at each possible price during a given period of time.

Market DEMAND

- ⇒ Refer to the quantity of a commodity that **all consumers are** willing and a buy, at each possible price during a given period of time.

Determinants of Demand

1. **Price of the given commodity:** There exists an **inverse relationship** between price and quantity demanded. It means, as price increase, quantity demanded falls due to decrease in the satisfaction level of consumers,
2. **Price of related goods:** Demand for the given commodity is also affected by change in prices of the related goods.

Related goods are two types:

- a. **Substitute goods:** Substitute goods are those goods which can be used in place of one another for satisfaction of a particular want, **like tea and coffee**.

An increase in the price of substitute leads to an increase in the demand for given commodity and vice-versa.

- b. **Complementary goods:** Complementary goods are those goods which are used together to satisfy a particular want, **like tea and sugar**.

An increase in the price of complementary goods to a decrease in the demand for given commodity and vice-versa

3. **Income of the consumer:** The effect of change in income on demand depends on the nature of the commodity under consideration.

- If the given commodity is a **normal good**, then an increase in income leads to **rise in this demand**, while a **decrease in income reduces the demand**.
- If the given commodity is an **inferior good**, then an **increases reduces the demand**, while a **decrease in income leads to rise in demand**.
- **For examples:** Suppose, income of a consumer increase. As a result, the consumer reduces consumption of toned milk and increase consumption of full cream milk. In this case, "toned milk" is an inferior good and "full cream milk" is a normal good.

4. **Tastes and preferences:** Tastes and preference of the consumer directly influence the demand for a commodity
5. **Expectation of change in the price in future:** If the price of a certain commodity is expected to increase in near future, then people will buy more of that commodity than what they normally buy. There exists a direct relationship between expectation of change in the prices in future and change in demand in the cent period.

Law of Demand

⇒ **Law of Demand:** Law of demand states the inverse relationship between price and quantity demanded, keeping other factors constant (*ceteris paribus*).

- Price ↑ Quality demanded ↓
- Price ↓ Quality demanded ↑
- **Ceteris paribus:** Other things being constant or unchanged.

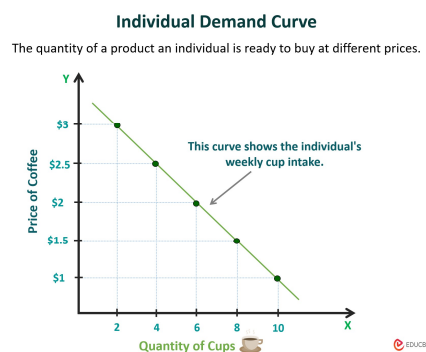
Demand Schedule

⇒ The demand schedule is a table that shows the relationship between the price of the good and the quantity demanded.

Demand Curve

⇒ The demand curve is a graphical representation of the demand schedule.

1. Individual Demand Curve:



2. Market Demand Curve:

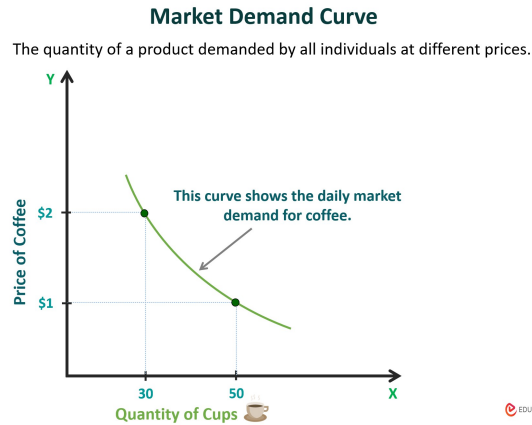


Fig1

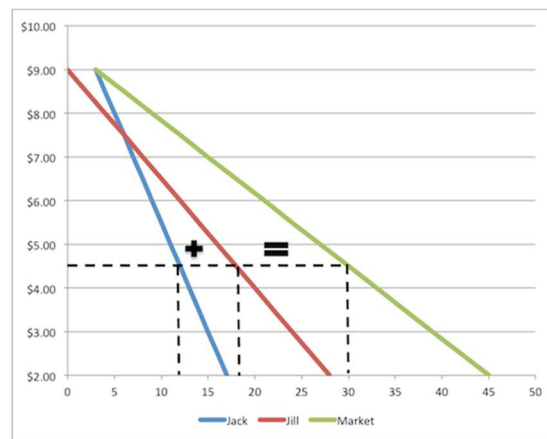


Fig 2

Shift in Demand Curve

- Increase in Demand:** Refers to a rise in demand of a commodity caused due to any factor other than the own price of the commodity.

Price (TK)	Demand (Units)
30	100
30	150



Fig: Increases in Demand,
Rightward Shift

2. **Decrease in Demand:** Refers to a fall in the demand of a commodity caused due to any factor other than the own price of the commodity

Price (TK)	Demand (Units)
30	100
30	80

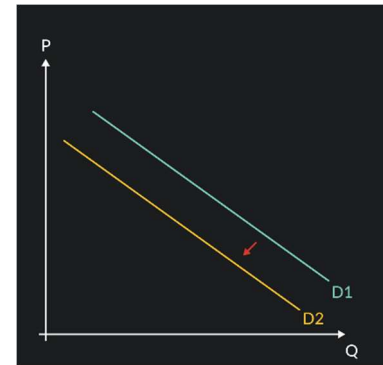


Fig: Decrease in Demand, Leftward Shift

Movement in Demand Curve

1. **Expansion in Demand:** Expansion in Demand Refers to a **rise in** the quantity demanded due to a **fall in the price of commodity**, other factors remaining Constant.

- It leads to a **downward movement** along the same demand curve.
- Also known as "Extension in Demand" or "**Increase in Quantity Demanded**"

Price (TK)	Demand (Units)
30	100
15	150

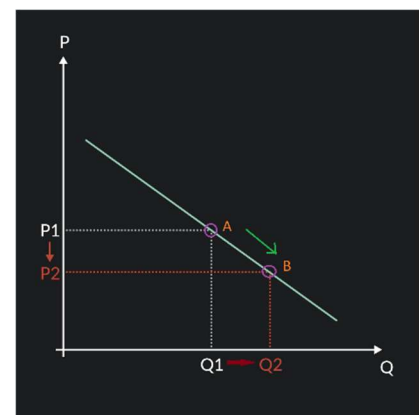


Fig: Increases in Demand, downward movement

2. Contraction in Demand:-Refers to a **fall** in the quantity demanded due to a **rise** in the price of commodity, other factors remaining constant.

- It leads to an **Upward movement** along the same demand curve.
- Also known as "Contraction in Demand" or "**Decrease in Quantity Demanded**"

Price (TK)	Demand (Units)
30	100
50	50

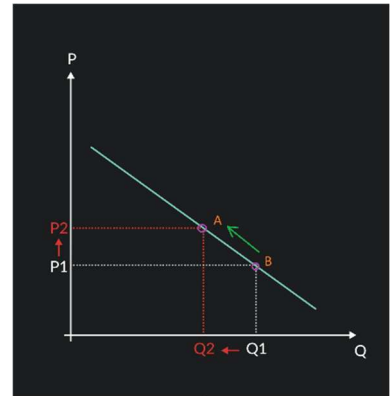


Fig: Decrease in Demand,
Upward movement

Reasons for Shift in Demand Curve

- Change in price of substitute goods.
- Change in price of complementary goods.
- Change in income of consumers.
- Change in tastes and preferences.
- Expectation of change in price in future.
- Change in population.
- Change in distribution of income.
- Change in season.

Reasons for Movement in Demand Curve

- **Extension of demand** (increase in quantity demanded) when the price falls.
- **Contraction of demand** (decrease in quantity demanded) when the price rises.

SYPPLY

⇒ Supply refers to **quantity of a commodity** that a firm is **willing** and **able to** offer for sale at a given **price** during a given **period of time**

There are Two types:

3. Individual Supply.

4. Market Supply.

Individual Supply

⇒ Refer to the quantity of a commodity that **an individual firm** is willing and able to offer for sale at a given **price** during a given **period of time**

Market Supply

⇒ Refer to the quantity of a commodity that **all firm** are willing and able to offer for sale at a given **price** during a given **period of time**

Determinants of Supply

1. **Price of the given commodity:** Price of a commodity and its supply are **directly** related. It means, as **price increases**, the quantity supplied of the given commodity **also rises** and vice-versa.

Price ↑ Supply ↑

2. **Price of other goods:**

Let, a supplier supply two goods Q1, Q2 at price P1, P2. Now if the price of Q1 increase the supplier increase the supply of Q1 goods and decrease the supply of Q2 goods and vice-versa.

3. Price of the factors of production or inputs: Price of the factors of production or inputs (like, labor, capital, raw material, etc.) used in the process of production constitute the cost of production of the commodity.

- a) If the prices of all or any of these factors or inputs **increases, the cost of production also increases**. This **decreases the profitability**. As a result, **seller reduces the supply of the commodity**.
- b) **Decreases** in prices of factors of production or inputs, **increases the supply** due to fall in cost of production and subsequent **rise in profit margin**.

4. States of Technology:

- a) Technological changes influence the supply of a commodity. **Advanced and improved technology reduces the cost** of production, which **raises the profit margin**. It induces **a seller to increase the supply**.
- b) Technological **degradation** or **complex** and **outdated technology** will **increase the cost of production** and it will lead to **decrease in supply**.

5. Expectation of change in the price in future: If sellers expect a rise in price in near future, then current market supply will decrease in order to raise the supply in future at higher prices. However, if the sellers fear that the prices will fall in the future, then they will increase the present supply to avoid losses in future.

6. Government Policy (Taxes & Subsidies): Increase in taxes raises the cost of production and, thus, a reduces the supply, due to lower profit margin. On the other hand, tax concessions increases supply as they make it more profitable for the firms to supply goods.

7. Means of transportation and communication: Proper infrastructural development, like improvement in the means of transportation and communication, help in maintaining adequate supply of the commodity.

Law of Supply

⇒ The law of supply states that, other things equal, the quantity supplied of a good, rises when the price of the good rises.

- Price↑ Supply↑
- Price ↓ Supply ↓
- **Ceteris paribus**: Other things being constant or unchanged.

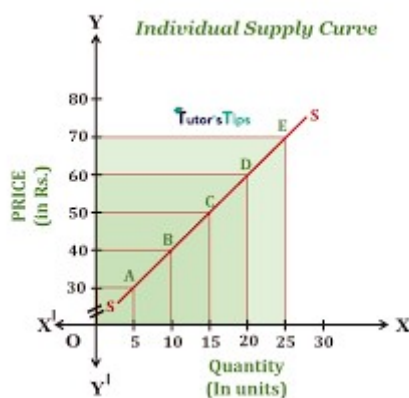
Supply Schedule

⇒ The Supply schedule is a table that shows the relationship between the price of the good and the quantity **supplied**.

Supply Curve

⇒ The Supply curve is a graphical representation of the Supply schedule.

1. **Individual Supply Curve:**



2. **Market Supply Curve:**



2. Market Supply Curve:

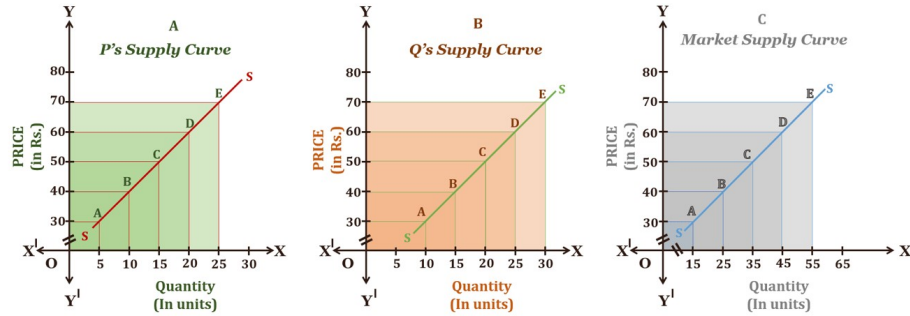


Fig 2

Shift in Supply Curve

- 1. Increase in Supply:** Refers to a rise in supply of a commodity caused due to any factor other than the own price of the commodity.

Price (TK)	Quantity (Units)
5	100
5	200

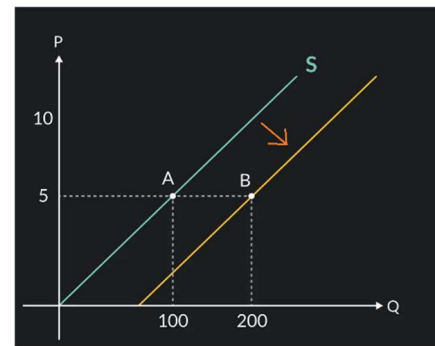


Fig: Increases in Supply,
Rightward Shift

- 2. Decrease in Supply:** Refers to a fall in the Supply of a commodity caused due to any factor other than the own price of the commodity

Price (TK)	Quantity (Units)
5	100
5	25

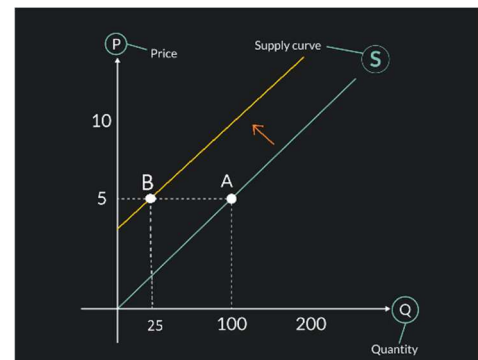


Fig: Decrease in Supply,
Leftward Shift

Movement in Supply Curve

1. Expansion in Supply: Expansion in Supply Refers to a **rise in** the quantity demanded due to a **increase in the price of commodity**, other factors remaining Constant.

- It leads to a **upward movement** along the same supply curve.
- Also known as "Extension in Supply" or "**Increase in Quantity Supplied**"

Price (TK)	Quantity (Units)
5	100
10	200

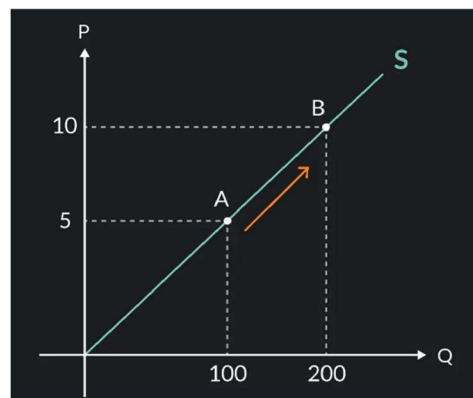


Fig: Increases in Supply,
Upward movement

2. Contraction in Supply:-Refers to a **fall in** the quantity Supplied due to a **fall in the price** of commodity, other factors remaining constant.

- It leads to an **Downward movement** along the same Supply curve.
- Also known as " Contraction in Supply" or "**Decrease in Quantity Supplied.**"

Price (TK)	Demand (Units)
5	100
2.5	50

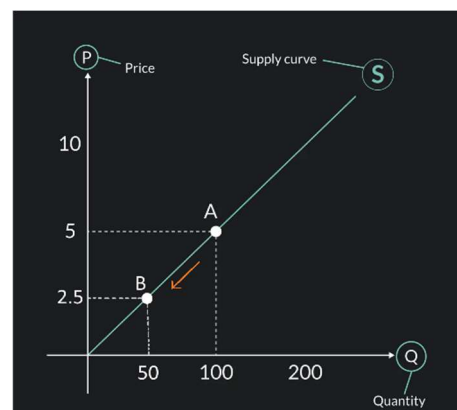


Fig: Decrease in Supply,
Downward movement

Reasons for Shift in Supply Curve

- **Changes in production costs** (e.g., labor, raw materials).
- **Technological advancements** improving efficiency.
- **Taxes and subsidies** affecting production costs.
- **Number of suppliers** entering or exiting the market.
- **Natural events or disasters** disrupting supply.
- **Government regulations** increasing or decreasing production constraints.
- **Price expectations** of future goods.
- **Availability of resources** or inputs.

Reasons for Movement in Supply Curve

- **Extension of supply** (increase in quantity supplied) when the price rises.
- **Contraction of supply** (decrease in quantity supplied) when the price falls.

SUPPLY AND DEMAND TOGETHER (*Equilibrium*)

⇒ **Equilibrium** refers to a situation in which the price has reached the level where **quantity supplied equals quantity demanded**.

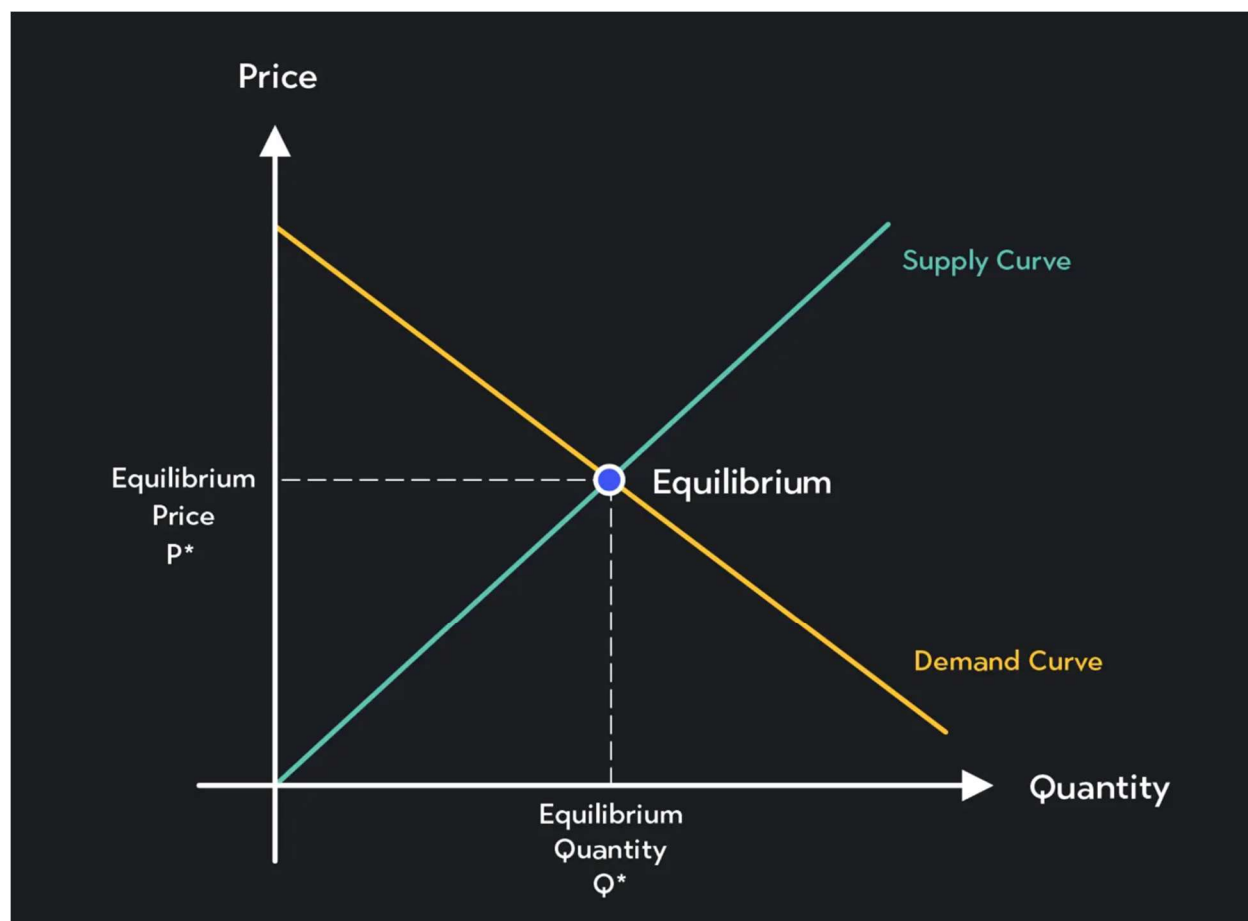
Equilibrium Price

- The price that balances quantity supplied and quantity demanded.
- On a graph, it is the price at which the **supply and demand curves intersect**.

Equilibrium Quantity

- The quantity supplied and the quantity demanded at the equilibrium price.
- On a graph it is the quantity at which the **supply and demand curves intersect**.

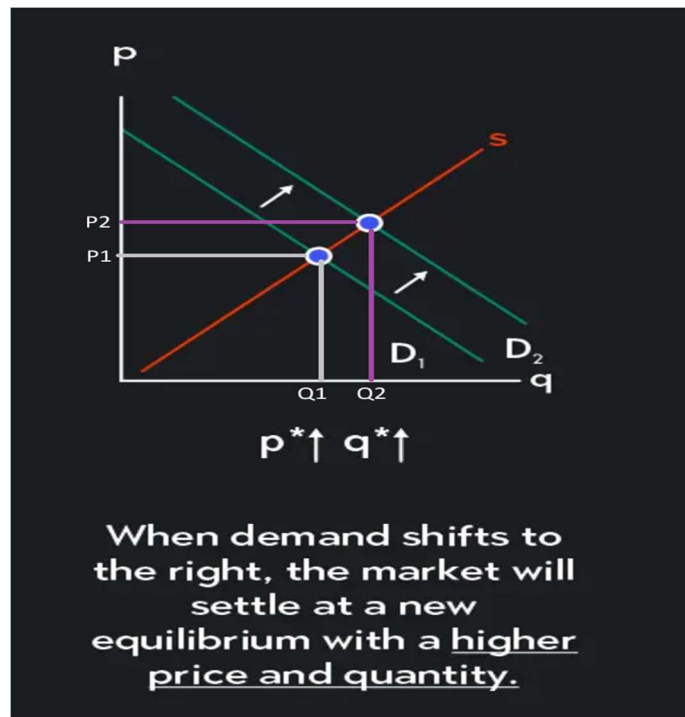
The Equilibrium of Supply and Demand



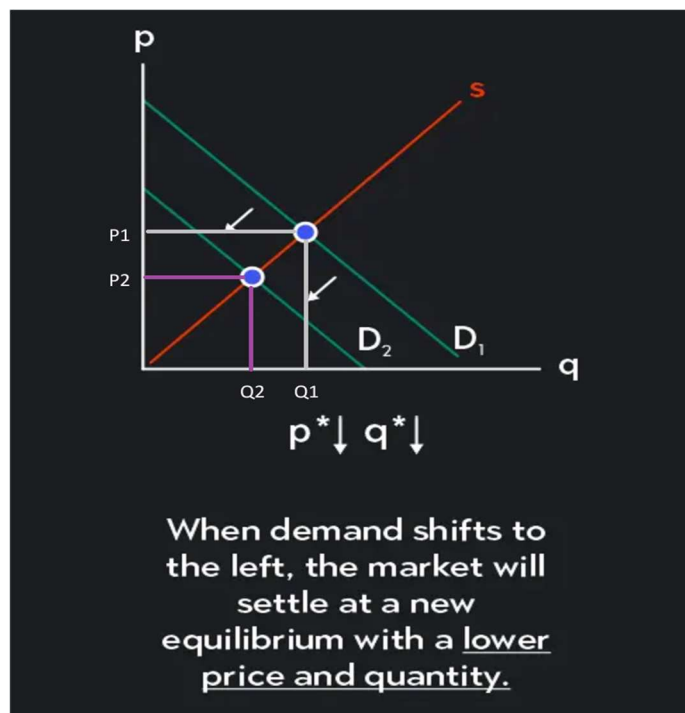
Steps To Analyzing Changes in Equilibrium

1. Decide whether the event shifts the supply or demand curve (or both).
2. Decide whether the curve(s) shift(s) to the left or to the right.
3. Use the supply-and-demand diagram to see how the shift affects equilibrium price and quantity.

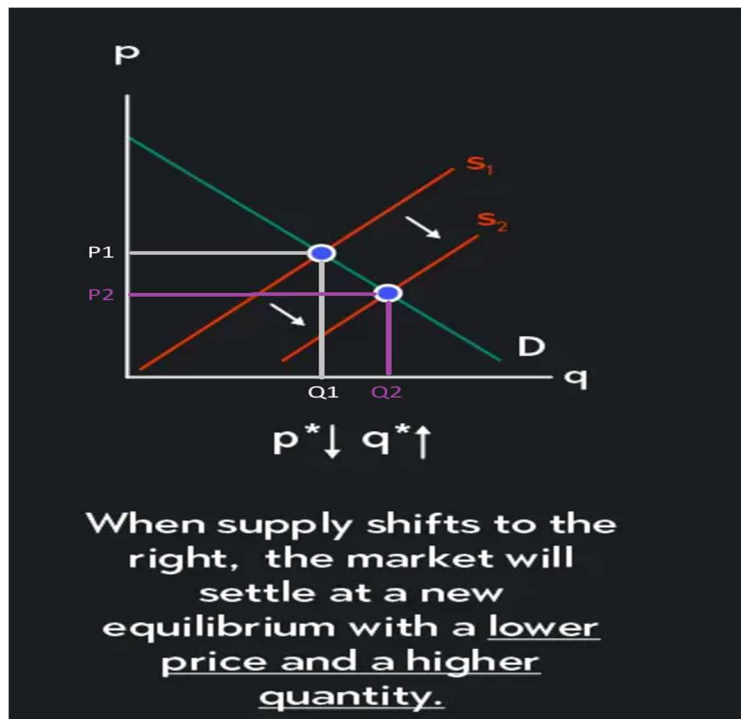
How an Increase Demand Affects the Equilibrium



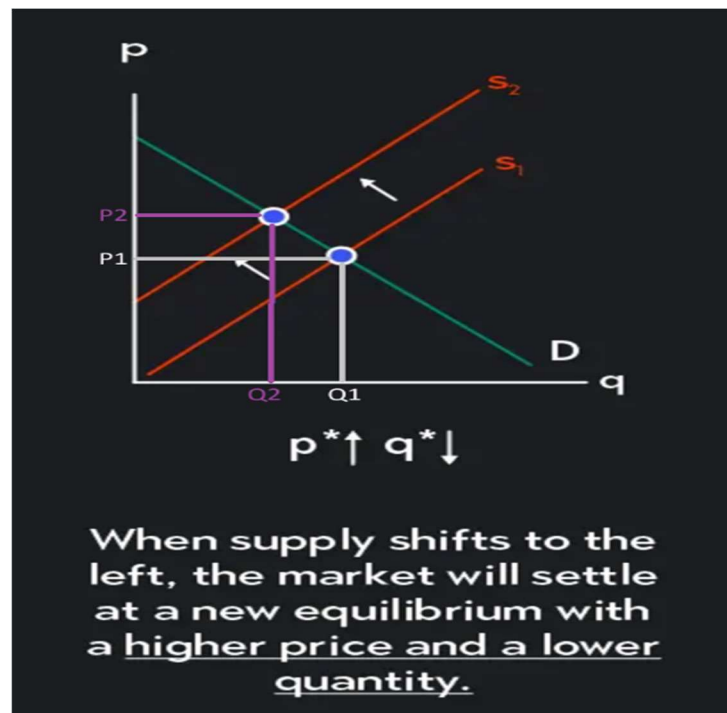
How a Decrease Demand Affects the Equilibrium



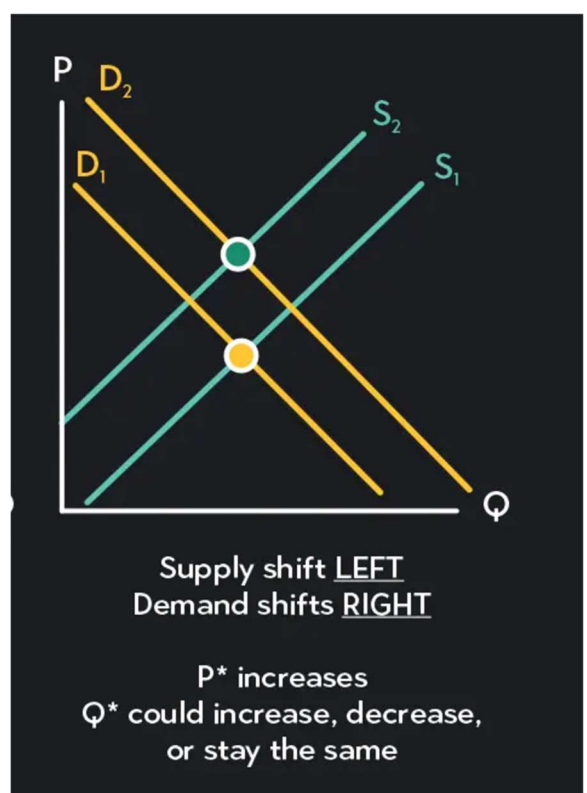
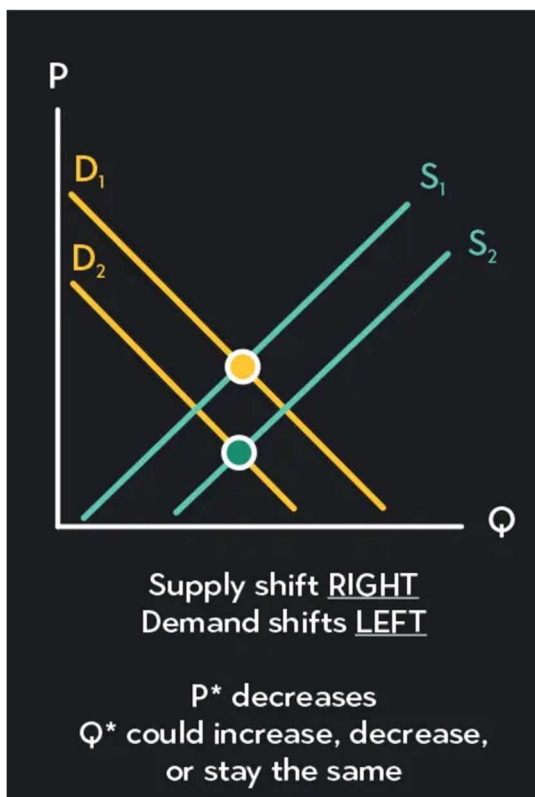
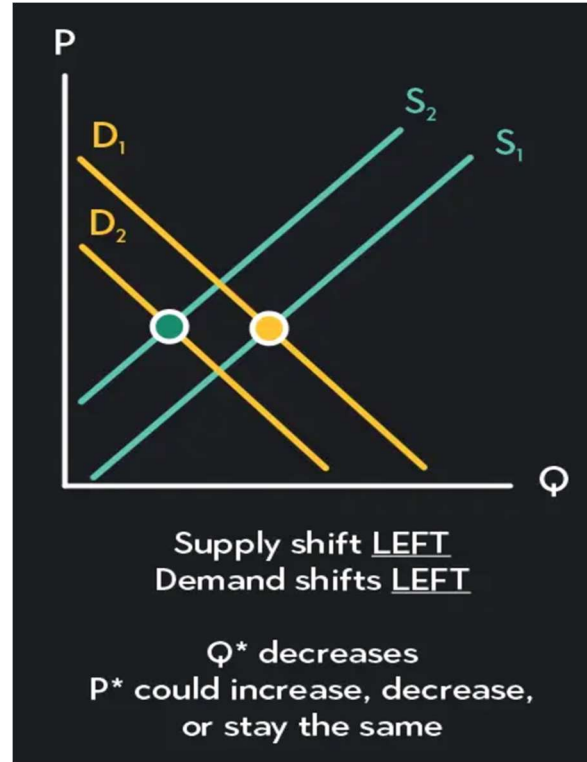
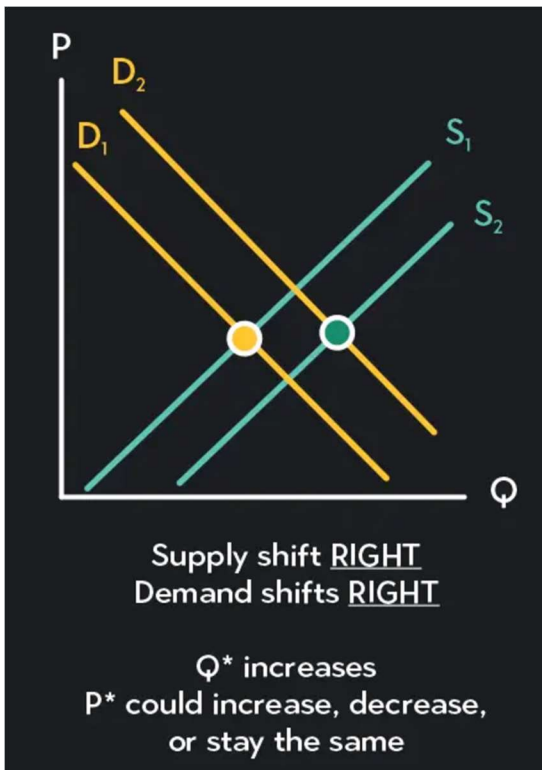
How an Increase Supply Affects the Equilibrium



How a Decrease Supply Affects the Equilibrium



A Shift in Both Supply and Demand



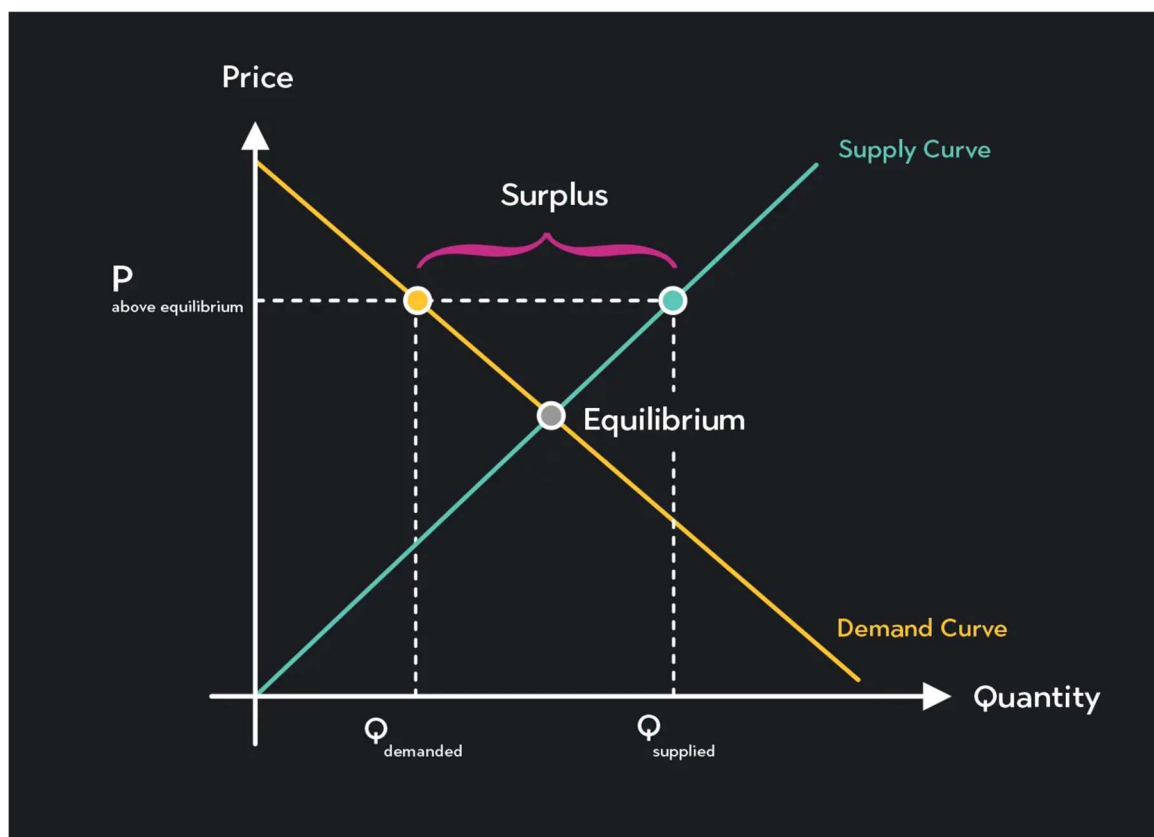
What Happens to Price and Quantity when Supply or Demand Shifts

	No Change in Supply	An Increase in Supply	A Decrease in Supply
No Change in Demand	P same Q same	P down Q up	P up Q down
An Increase in Demand	P up Q up	P ambiguous Q up	P up Q ambiguous
A Decrease in Demand	P down Q down	P down P ambiguous	P ambiguous Q down

Surpluses and Shortages

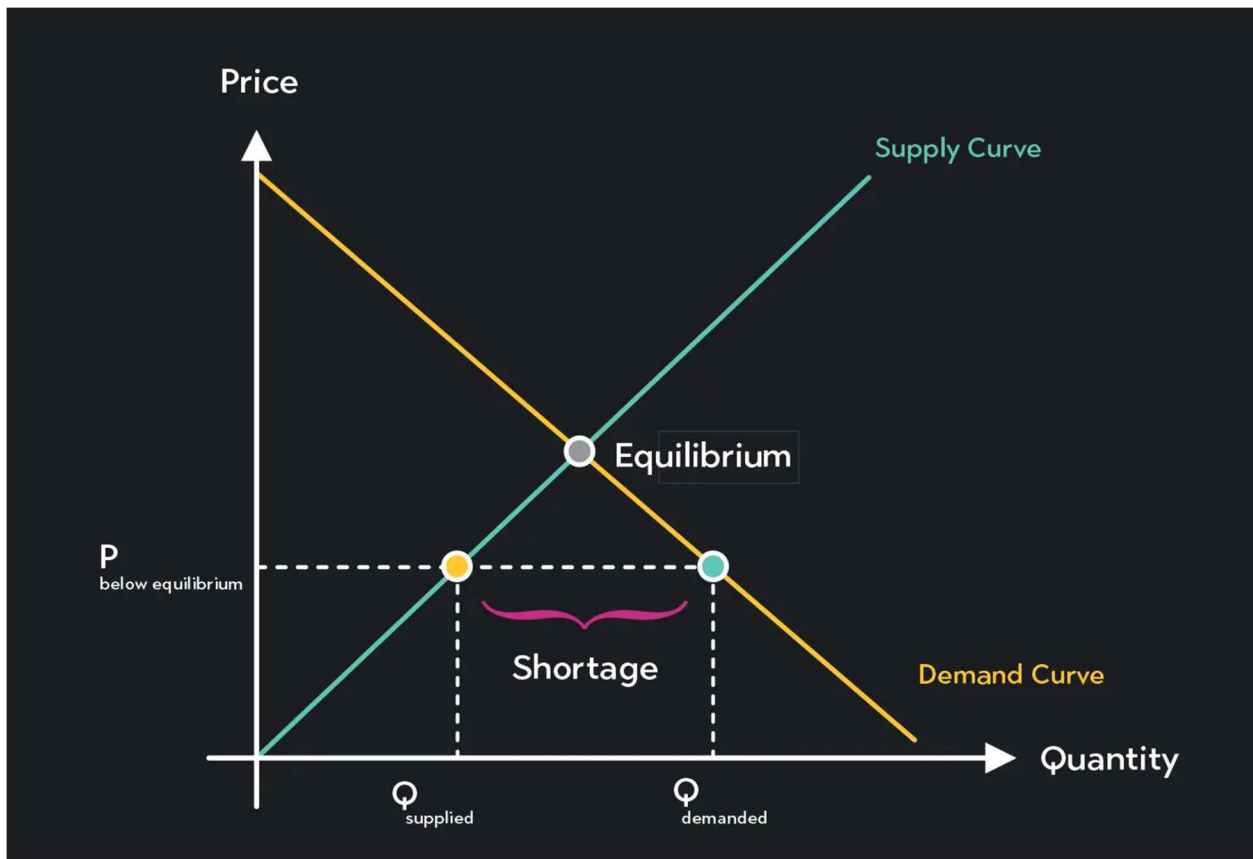
• **Surplus**

- ⇒ When price > equilibrium price, then quantity supplied > quantity demanded.
- ⇒ There is excess supply or a surplus.
- ⇒ Suppliers will lower the price to increase sales, thereby moving toward equilibrium.



- **Shortage**

- ⇒ When price < equilibrium price, then quantity demanded > the quantity supplied.
- ⇒ There is excess demand or a shortage.
- ⇒ Suppliers will raise the price due to too many buyers chasing too few goods, thereby moving toward equilibrium.



#Problem 1:

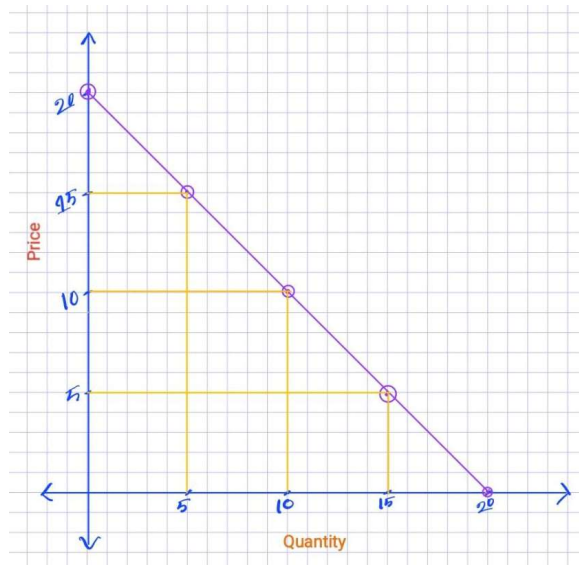
1. Following Table shows the demand schedule of Good X.

Price of X	Quantity demanded of X
20	0
15	5
10	10
5	15
0	20

- Draw the demand curve for Good X.
- Suppose Good Y is a substitute for good X. On such case what will be the change in the demand curve of good X, Show that.

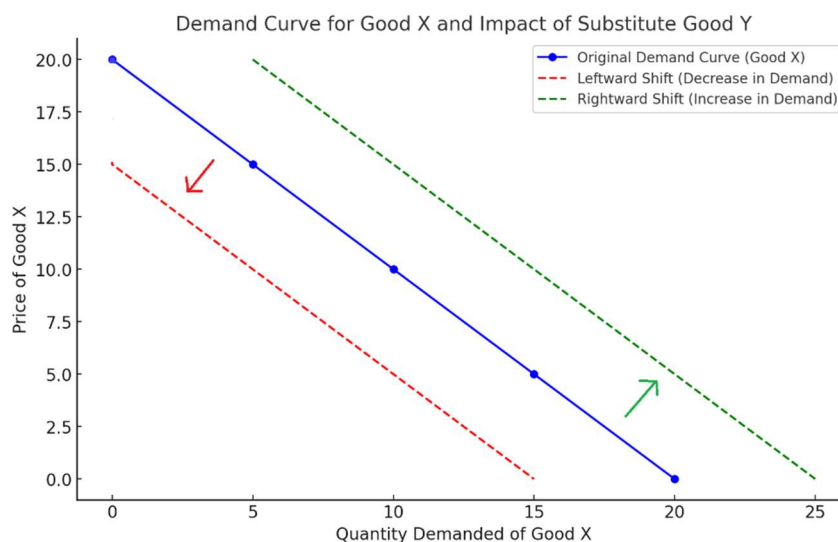
(a)

Now the Demand curve is:



(b)

- If the price of Good Y decreases, consumers may shift to Good Y, reducing the demand for Good X. This causes the **demand curve for Good X to shift leftward**.
- Conversely, if the price of Good Y increases, consumers may switch to Good X, increasing its demand. This causes the **demand curve for Good X to shift rightward**.



#Problem 2:

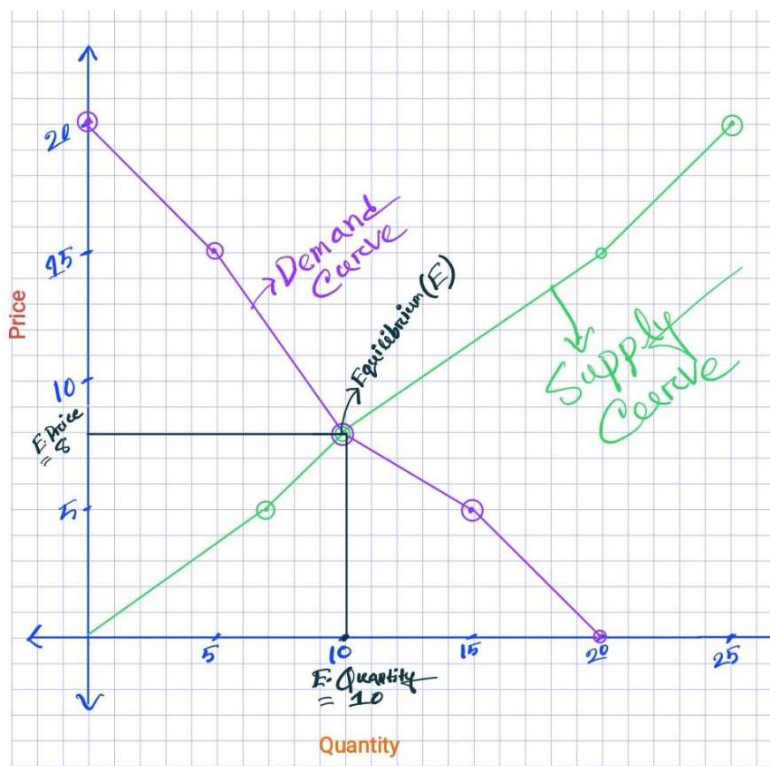
2. Following Table shows the supply and demand schedule of sweater.

Price	Quantity Demanded	Quantity Supply
20	0	25
15	5	20
8	10	10
5	15	7
0	20	0

- Draw the demand and supply curve for Good X.
- Identify the equilibrium point and what is the equilibrium price and quantity in the sweater market?
- Suppose in winter season the weather is **extremely** cold. In addition to that due to extreme cold **some** of the sheep got died which is the important resources for sweater production. How do these two events affect the market equilibrium (demand and supply). How will it change the equilibrium price and quantity?

(a)

Now the Demand & Supply curve is:



(b)

We Know,

The **equilibrium** is the point where $Q_D = Q_S$

From the table, at $P=8$ $Q_D = Q_S = 10$.

- **Equilibrium Price:** $P=8$
- **Equilibrium Quantity:** $Q=10$

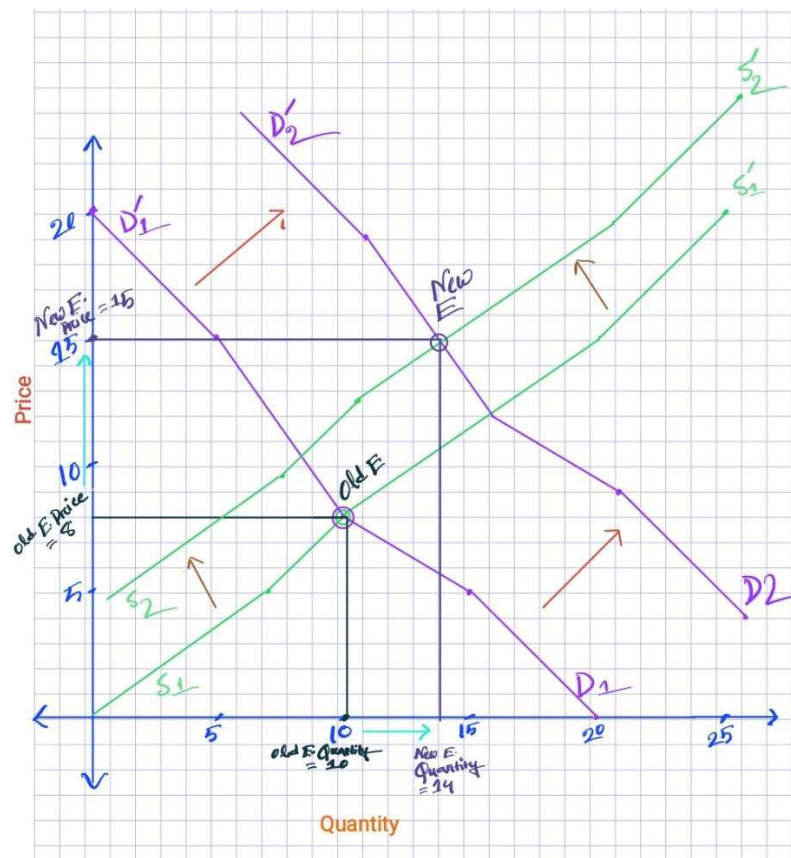
(c)

➔ **For Extreme cold weather (increase in demand):**

- Increased demand for sweaters as more people needs them for warmth.
- The **demand curve shifts rightward**.

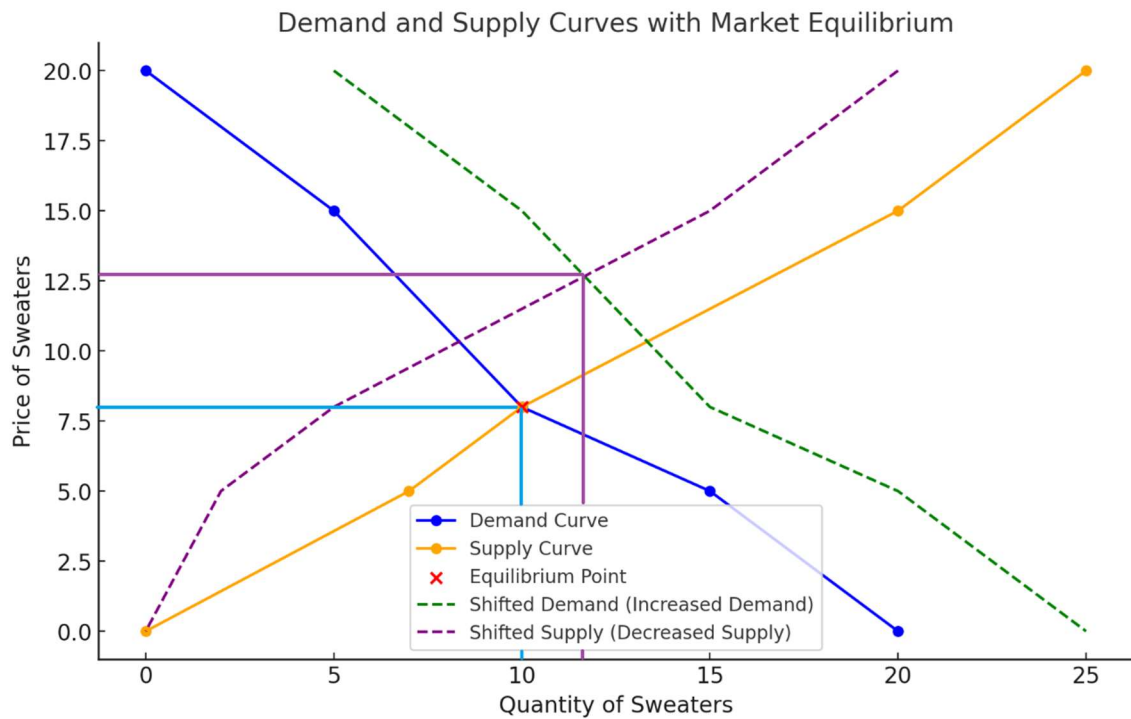
➔ **For Reduction in sheep population (decrease in supply):**

- Fewer resources for sweater production.
- The **supply curve shifts leftward**.



Effects on the Equilibrium:

- The **equilibrium price** increases due to the combined effects of higher demand and lower supply.
- The **equilibrium quantity** also increases but not so much due to reduced production capacity, despite increased demand.



ELASTICITY AND ITS APPLICATIONS

TOPIC: 3

THE ELASTICITY OF DEMAND

⇒ It is a measure of how much buyers respond to changes in market conditions. **allows us to analyze demand with greater precision.**

Four Types of Elasticity:

1. Price Elasticity of Demand (PED)
2. Cross Price Elasticity of Demand (XED)
3. Income Elasticity of Demand (YED)
4. Advertising Elasticity of Demand (AED)

Price Elasticity of Demand(PED)

- ⇒ *Price elasticity of demand* is a measure of how much the quantity demanded of a good respond to a change in the price of that good.
- ⇒ Price elasticity of demand is the percentage change in quantity demanded given a percent change in the price.

$$\text{PED} = \% \text{ change in quantity} / \% \text{ change in price}$$

Or

$$\text{PED} = [(Q_2 - Q_1) / Q_1] / [(P_2 - P_1) / P_1]$$

Q1 = initial quantity of demand

Q2 = new quantity of demand

P1 = initial price

P2 = new price

Cross Price Elasticity of Demand (XED):

- ⇒ Cross price elasticity of demand (XED) measures how changes in the price of one product affect the demand for another, applicable to related goods: **substitutes** or **complements**.
- **Substitutes:** A price increase in one product leads to higher demand for the other (e.g., coffee and tea).
 - **Complements:** A price increase in one product leads to lower demand for the other (e.g., coffee and creamer).
 - **Unrelated Goods:** $XED = 0$ (e.g., airline tickets and oranges).

XED = % change in quantity for product A / % change in price for product B

Or

$$\text{XED} = [(Q_{2a} - Q_{1a}) / (Q_{2a} + Q_{1a})] / [(P_{2b} - P_{1b}) / (P_{2b} + P_{1b})]$$

Q_{1a} = initial quantity of demand of product A

Q_{2a} = new quantity of demand of product A

P_{1b} = initial price of product B

P_{2b} = new price of product B

Income Elasticity of Demand (YED):

⇒ Income Elasticity of Demand (YED) measures how demand changes with customer income (Y is the notation for income).

- **Normal Goods:** Demand increases as income rises (e.g., food, clothes).
- **Luxury Goods:** Demand rises more significantly but lags behind income increases.
- **Inferior Goods:** Demand decreases as income rises.

YED = % change in quantity / % change in income

Or

$$\text{YED} = [(Q_2 - Q_1) / Q_1] / [(Y_2 - Y_1) / Y_1]$$

Q_1 = initial quantity of demand

Q_2 = new quantity of demand

Y_1 = initial income

Y_2 = new income

Advertising Elasticity of Demand (AED):

⇒ Advertising Elasticity of Demand (AED) measures how changes in advertising affect customer demand, evaluating advertising effectiveness in driving sales.

AED = % change in quantity / % change in advertising

Or

$$\text{AED} = [(Q_2 - Q_1) / Q_1] / [(A_2 - A_1) / A_1]$$

Q_1 = initial quantity of demand

Q_2 = new quantity of demand

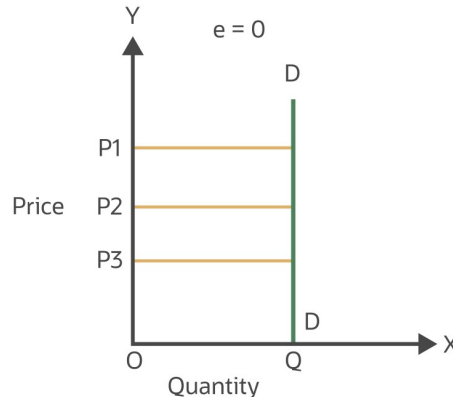
A_1 = initial advertising expenditure

A_2 = new advertising expenditure

Five Categories of Elasticity of Demand

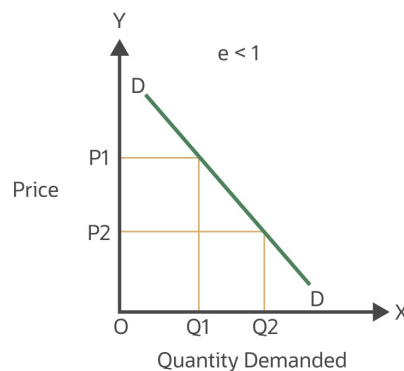
1. Perfectly inelastic demand:

- ⇒ It occurs when **demand remains unchanged** regardless of **changes in other factors, such as price**. This typically applies to necessities with no substitutes (e.g., essential foods, lifesaving medicine). **PED=0**.



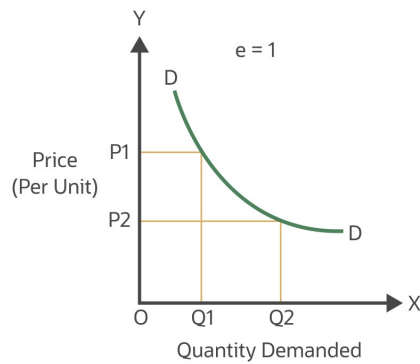
2. Relatively inelastic demand:

- ⇒ It occurs when **large changes in factors (e.g., price)** cause only **small changes in demand**. Common examples include gasoline and salt. **PED < 1**.



3. Unitary elastic demand:

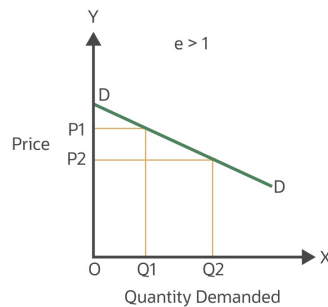
- ⇒ It occurs when **changes in factors (e.g., price)** result in **proportional changes in demand**. For instance, a 10% price increase causes a 10% decrease in demand. **PED=1**.



4. Relatively elastic demand:

⇒ It occurs when a **small change in one factor (e.g., price)** leads to a **disproportionately larger change in demand**.

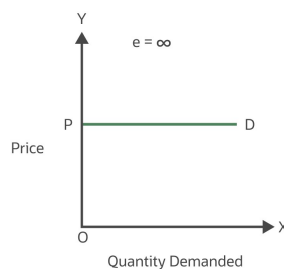
- **Example:** A 5% price increase in a streaming service causing a 10% drop in subscribers.
- **PED > 1.**
- Higher PED values indicate greater elasticity.



5. Perfectly elastic demand:

⇒ It is an extreme case where **demand drops to zero with any change in factors, such as price**.

- **Example:** If organic bananas become more expensive at one store but remain the same at another, all demand shifts to the cheaper store.
- **PED = ∞**



Example 1:

→ KMR Inc. is in the online retail shoe business. In 2021, KMR sold **1,500 pairs of snow boots** at an average price of **\$100 per pair**. During 2022, KMR lowered the price to **\$90** and sold **1,800 pairs**. In both years, KMR's cost of goods sold was \$40. Calculate the price elasticity of demand?

PED = % change in quantity / % change in price

$$\text{PED} = [(Q_2 - Q_1) / Q_1] / [(P_2 - P_1) / P_1]$$

$$Q_1 = 1,500$$

$$Q_2 = 1,800$$

$$P_1 = \$100$$

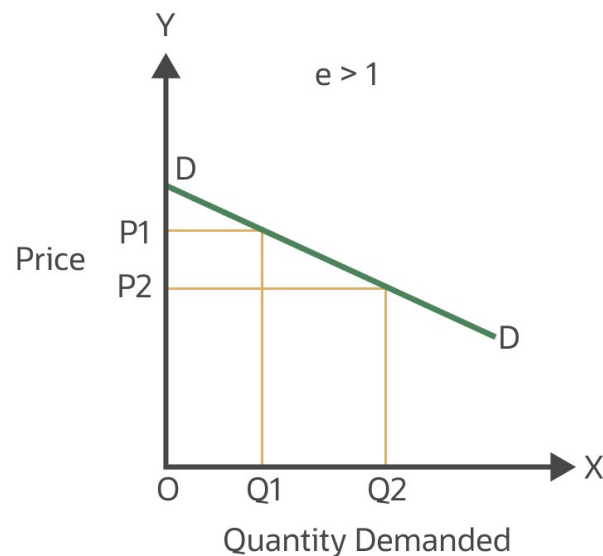
$$P_2 = \$90$$

$$= [(1,800 - 1,500) / 1,500] / [(\$90 - \$100) / \$100]$$

$$= 0.2 / 0.1$$

$$= 2$$

Since, **PED > 1** that means it is **Relatively elastic demand**. It occurs when a **small change** in one factor (e.g., price) leads to a **disproportionately** larger change in demand.



The Midpoint Method for PED

- ⇒ A Better Way to Calculate Percentage Changes and Elasticities.
- ⇒ Price elasticity of demand is the percentage change in quantity demanded given a percent change in the price.

Midpoint Formula for PED

$$\text{Price Elasticity of Demand} = \frac{\frac{\text{Change in Quantity Demanded}}{\text{Average Quantity Demanded}}}{\frac{\text{Change in Price}}{\text{Average Price}}}$$
$$\text{PED} = \frac{\frac{Q_2 - Q_1}{(Q_1 + Q_2)/2}}{\frac{P_2 - P_1}{(P_1 + P_2)/2}}$$

Q1 = initial quantity of demand

Q2 = new quantity of demand

P1 = initial price

P2 = new price

Example 2:

→ Suppose the price of a T-shirt is \$14, and the quantity demanded of this T-shirt is 60 units. But after an increase in the price of the T-shirt from \$14 to \$16, the quantity demanded for this T-shirt decreased from 60 units to 40 units. Calculate the price elasticity of demand use the midpoint formula?

→ Is given,

$$P_1=14, P_2=16$$

$$Q_1=60, Q_2=40$$

$$\text{Now average price} = \frac{P_1 + P_2}{2} = \frac{14 + 16}{2} = 15$$

$$\text{Now average Quantity} = \frac{Q_1 + Q_2}{2} = \frac{60 + 40}{2} = 50$$

$$\text{We Know, PED} = \frac{\frac{Q_2 - Q_1}{\text{average Quantity}}}{\frac{P_2 - P_1}{\text{average price}}} = \frac{\frac{40 - 60}{50}}{\frac{16 - 14}{15}} = -3$$

- ⇒ The **negative sign** shows that there is an **inverse relationship** between the price of T-shirt and its quantity demanded, according to the law of demand. The price and the quantity demanded move in opposite directions.
- ⇒ The magnitude of PED is 3 which is **greater than 1**, indicating that the demand for T-shirts is **elastic** and consumers are responsive to the change in the price of T-shirts.

Interpretation of Income Elasticity of demand

- ⇒ When the value of **Income elasticity of demand** is **negative** then it is considered to have **inferior goods** in the market.
- ⇒ When the value of **Income elasticity of demand** is **positive** then it is considered to have **normal goods** in the market.
 - When **IED or YED** is between **0 and <1** then it is **income inelastic**
 - When **IED or YED >1** then it is **income elastic**
 - When **IED or YED =1** then it is **income unit elastic**

Interpretation of Cross Price Elasticity of demand

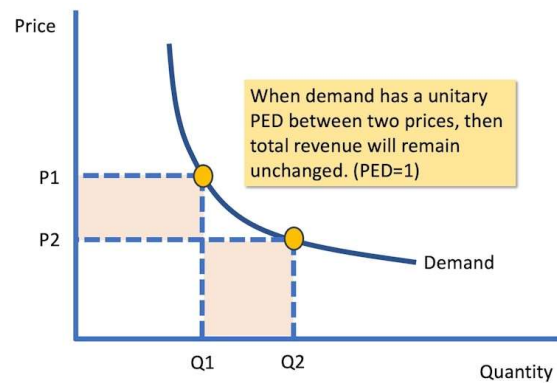
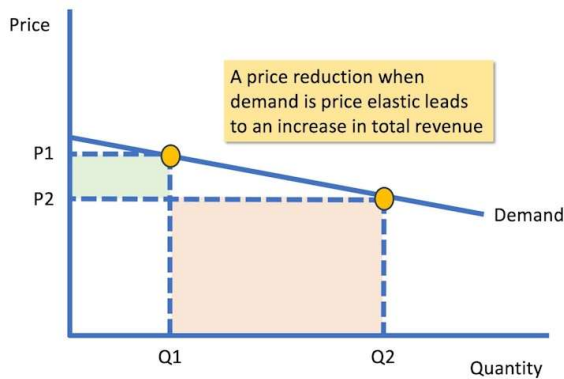
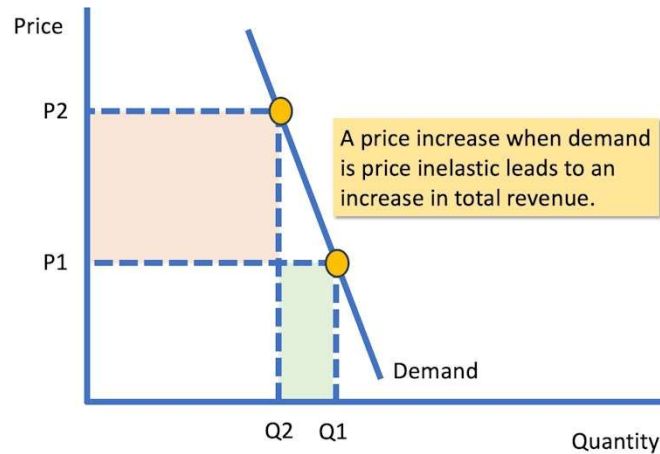
- ⇒ When the value of **Cross price elasticity of demand** is **negative** then it is considered to have **complementary goods** in the market.
- ⇒ When the value of **Cross price elasticity of demand** is **positive** then it is considered to have **substitute goods** in the market.
 - When **XED** is between **0 and <1** then it is **XED inelastic**
 - When **XED >1** then it is **XED elastic**
 - When **XED =1** then it is **XED unit elastic**

Elasticity and Revenue

- ⇒ A firm's total revenue is equal to the number of products it sells times the price of the product.

$$\begin{array}{c} \text{Total Revenue} = \text{Price} \times \text{Quantity} \\ \text{or} \\ \text{TR} = P \times Q \end{array}$$

- ⇒ When a product is **elastic**, and its **price rises**, what happens to the firm's total revenue?
- ⇒ **For example**, if a store **sells 30 pairs of shoes at \$10 each**, then its **revenue equals 30 times \$10, or \$300**. If the store **sells 20 pairs of shoes after the price increases to \$25**, then its **total revenue equals 20 times \$25, or \$500**. So the store's **total revenue increases**.



⇒ In the above example, P (the price) **increased**, so, therefore, Q (the quantity demanded) **decreased**, and **total revenue increased**. Does a price increase always lead to total revenue increase? The answer is “no”. It depends on the product’s elasticity. Let’s look at the following example.

⇒ A supermarket **sells 50 oranges at \$1 each**. Its **revenue equals 50 times \$1 or \$50**. If the store **sells 20 oranges after the price increases to \$2**, then its **revenue equals 20 times \$2, or \$40**. In this case, the store’s **revenue decreases**.

Elastic Products:

- The percentage change in quantity demanded is greater than the percentage change in price ($PED > 1$).
- **Impact on Revenue:**
 - Price increase → Quantity demanded decreases significantly → Revenue decreases.
 - Price decrease → Quantity demanded increases significantly → Revenue increases.

Inelastic Products:

- The percentage change in quantity demanded is smaller than the percentage change in price ($PED < 1$).
- **Impact on Revenue:**
 - Price increase → Quantity demanded decreases slightly → Revenue increases.
 - Price decrease → Quantity demanded increases slightly → Revenue decreases.

⇒ In summary:

- When a product is elastic and its price falls, total revenue increases.
- When a product is elastic and its price rises, total revenue decreases.
- When a product is inelastic and its price rises, total revenue increases.
- When a product is inelastic and its price falls, total revenue decreases.
- When a product is unit elastic and its price changes, total revenue remains constant.

PRICE ELASTICITY OF SUPPLY

⇒ The Price elasticity of supply (PES) is a measure of the responsiveness of the quantity supplied of a product or service to the change in its price.

$$\text{PES} = \frac{\% \text{ Change in quantity supplied}}{\% \text{ Change in price}}$$

PRICE ELASTICITY OF SUPPLY FORMULA

Example 3.1:

→ Suppose an increase in the price of milk from \$1.90 to \$2.10 a liter raises the amount that dairy farmers produce from 9000 to 11 000 L per month. Calculate the price elasticity of Supply?

Is given,

$$\begin{aligned} P_1 &= 1.90, & P_2 &= 2.10 \\ Q_1 &= 9000, & Q_2 &= 11000 \end{aligned}$$

$$\text{We Know, } PES = \frac{\frac{Q_2 - Q_1}{Q_1}}{\frac{P_2 - P_1}{P_1}} = \frac{\frac{11000 - 9000}{9000}}{\frac{2.10 - 1.90}{1.90}} = 2.11$$

Example 3.2:

→ Suppose the price of a T-shirt is \$14, and the quantity demanded of this T-shirt is 60 units. But after an increase in the price of the T-shirt from \$14 to \$16, the quantity demanded for this T-shirt decreased from 60 units to 40 units. Calculate the price elasticity of demand use the midpoint formula?

→ Is given,

$$\begin{aligned} P_1 &= 1.90, & P_2 &= 2.10 \\ Q_1 &= 9000, & Q_2 &= 11000 \end{aligned}$$

$$\text{Now average price} = \frac{P_1 + P_2}{2} = \frac{2.10 + 1.90}{2} = 2$$

$$\text{Now average Quantity} = \frac{Q_1 + Q_2}{2} = \frac{11000 + 9000}{2} = 10000$$

$$\text{We Know, PES} = \frac{\frac{Q_2 - Q_1}{\text{average Quantity}}}{\frac{P_2 - P_1}{\text{average price}}} = \frac{\frac{11000 - 9000}{10000}}{\frac{2.10 - 1.90}{2}} = 2$$

Purpose of the Midpoint Method

- Precise Elasticity Measurement
- Eliminates Bias
- Consistent Comparison
- Suitable for Nonlinear Curves
- Widely Accepted

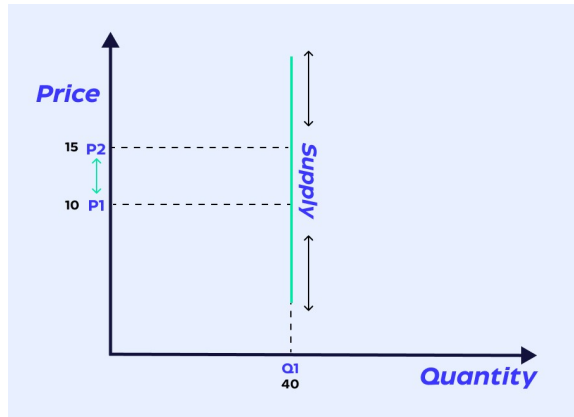
Disadvantages of the Midpoint Method

- Limited to Straight Lines
- Some Factors are Ignored
- Not Suitable More than Two Points

Types of Price Elasticity of Supply

1. Perfectly inelastic Supply:

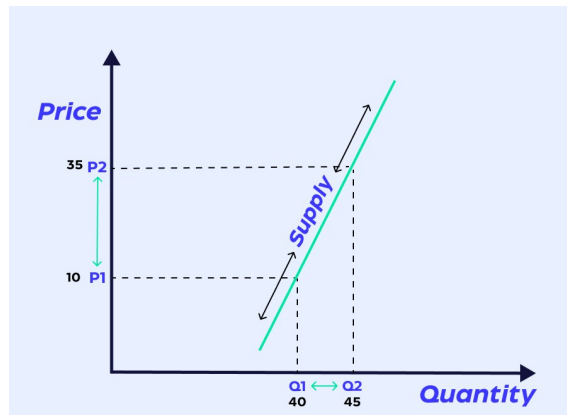
⇒ If price changes have **no effect on quantity supplied**, the supply is **perfectly inelastic**, meaning the supplier cannot adjust the quantity regardless of price changes. $PES=0$.



2. Relatively inelastic Supply:

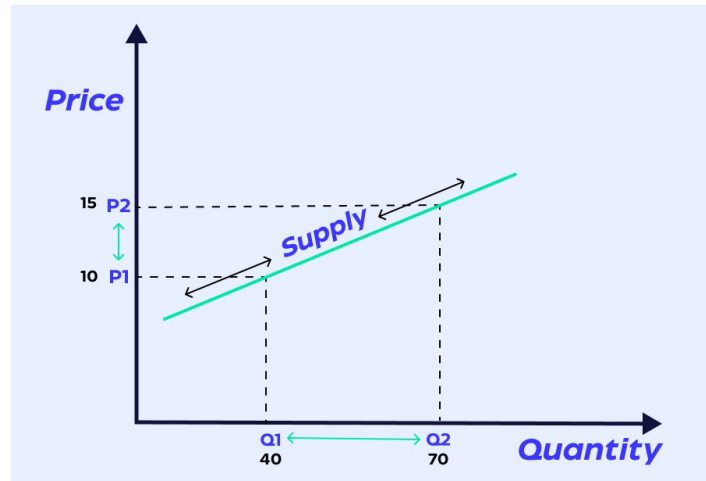
⇒ If the percentage change in **quantity supplied** is **less** than the **percentage change in price**, the supply is **relatively inelastic**, meaning suppliers are less responsive to price changes.

⇒ $PES < 1$.



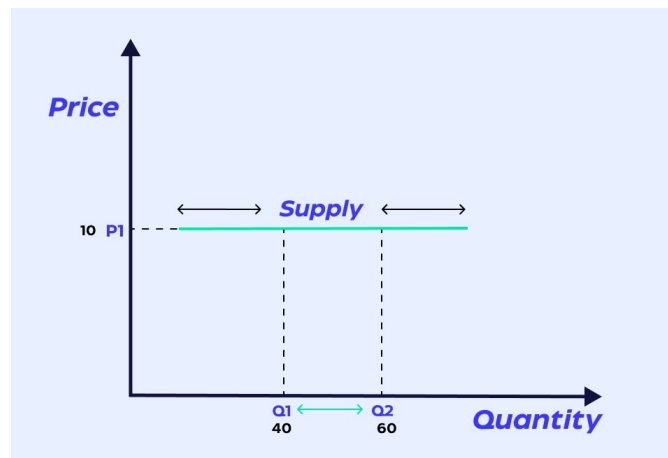
3. Relatively elastic Supply:

- ⇒ If the percentage change in **quantity supplied** is **greater** than the **percentage change in price**, the supply is considered **relatively elastic**, meaning the supplier is responsive to price fluctuations and can adjust the quantity supplied accordingly.
- ⇒ $PES > 1$.



4. Perfectly elastic Supply:

- ⇒ If a small price change causes an infinite change in quantity supplied, the supply is **perfectly elastic**, meaning suppliers can supply unlimited quantities at a specific price.
- ⇒ $PES = \infty$



#Problem 1:

⇒ Answer the following questions based on the table.

Point	Price of Shoes	Price of sandals	Demand for sandals	Demand for shoes
A	100	50	30	15
B	200	40	35	10
C	300	30	40	5
D	400	20	45	0

- a. Calculate price elasticity of demand for **shoes** from point A to C and interpret your answer.
- b. Calculate price elasticity of demand for **sandals** from point D to B and interpret your answer.
- c. Calculate cross price elasticity of demand **for shoes** from point B to C. Based on your answer identify whether the goods are substitutes or complements.

(a)

Is given,

$$\begin{array}{ll} P_1=100, & P_2=300 \\ Q_1=15, & Q_2=5 \end{array}$$

$$\text{We Know, } PED = \frac{\frac{Q_2 - Q_1}{Q_1}}{\frac{P_2 - P_1}{P_1}} = \frac{\frac{5 - 15}{15}}{\frac{300 - 100}{100}} = -0.333$$

Interpretation:

The price elasticity of demand for shoes from point A to C is **-0.33** (in absolute value). Since the magnitude is less than 1, this indicates that the demand for shoes is **Relatively inelastic**. This means that consumers are not very responsive to price changes in this range.

(b)

Is given,

$$\begin{array}{ll} P_1=20, & P_2=40 \\ Q_1=45, & Q_2=35 \end{array}$$

$$\text{We Know, } PED = \frac{\frac{Q_2 - Q_1}{Q_1}}{\frac{P_2 - P_1}{P_1}} = \frac{\frac{35 - 45}{45}}{\frac{40 - 20}{20}} = -0.222$$

Interpretation:

The price elasticity of demand for sandals from point D to B is **-0.22** (in absolute value). Since the magnitude is less than 1, this indicates that the demand for sandals is **Relatively inelastic**. The demand does not change much in response to the price change.

(c)

Is given,

$$\begin{array}{ll} P_{b1}=40, & P_{b2}=30 \\ Q_{a1}=10, & Q_{a2}=5 \end{array}$$

$$\text{We Know, } XED = \frac{\frac{Q_{a2}-Q_{a1}}{Q_{a1}+Q_{a2}}}{\frac{P_{b2}-P_{b1}}{P_{b1}+P_{b2}}} = \frac{\frac{5-10}{10+5}}{\frac{30-40}{40+30}} = 2.333$$

Interpretation:

The cross-price elasticity of demand for shoes from point B to C is **2.333**. Since the cross-price elasticity is positive, this indicates that shoes and sandals are **substitutes**. As the price of sandals decreases, the demand for shoes decreases, suggesting consumers shift their purchases between the two goods.

THE PRODUCTION & COSTS

TOPIC: 4

Cost, Profit, and Revenue

1. Revenue:

→ Revenue is money a firm brings in from sales. Total Revenue is Price times Quantity ($TR = P \times Q$).

2. Costs:

→ There are two types of costs we need to know in microeconomics. Explicit costs and implicit costs.

3. Profit

→ In microeconomics there are actually two types of profit we need to know: accounting profit and economic profit.

Opportunity Costs

⇒ **Economists like to say every choice has a cost. That cost is called an opportunity cost.** In economics, cost isn't just about money; it is about lost opportunities. For example, if you choose to go to soccer practice, you lose the opportunity to hang out with your friends. Hanging out with your friends is your opportunity cost. Opportunity cost is the value of the best alternative not chosen.

There are two types:

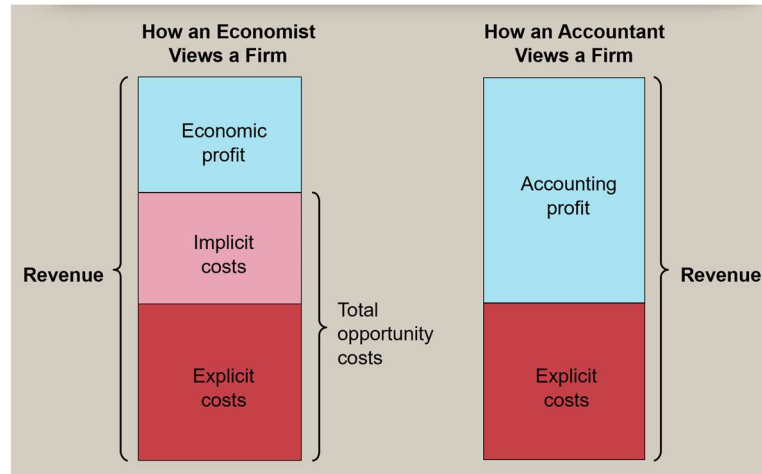
1. **Explicit Cost:** Direct monetary expenses, e.g., the price of a movie ticket, popcorn, and soda.
2. **Implicit Cost:** The value of lost opportunities, such as income foregone by taking a night off work to attend the movie.

$$\text{Opportunity cost} = \text{Explicit costs} + \text{Implicit costs.}$$

Economic Profit versus Accounting Profit

Accounting Profit: Total revenue minus explicit costs. It's always higher than economic profit and often tested in economics exams.

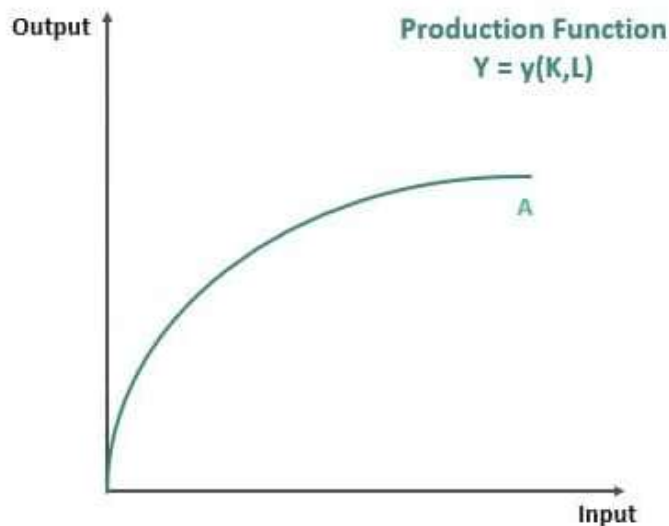
Economic Profit: Total revenue minus both explicit and implicit costs. More significant to economists, it reflects true profitability.



Production Function

⇒ Production and costs can be explained with the help of the production function. The production function is just the function relationship between the various production factors and the output produced through this process. Arithmetically this is depicted as $Q = f(L, K)$. The Q over here is the output, while 'L' and 'K' are the different inputs that go into the production process. There may be various other factors involved, but over here, we are working with a two-factor model.

Production Function Graph



Total, Average and Marginal Product

- ⇒ **Total Product** is the total amount of output produced in physical units such as bushels of wheat or number of sneakers.
- ⇒ **Marginal Product** of an input is the extra product or output produced by 1 additional unit of that input while other inputs are held constant.
For example, assume that we are holding land, machinery and all other inputs constant. Then labor's marginal product is the extra output obtained by adding 1 unit of labor.
- ⇒ **Average Product** is the total output divided by total units of input.
Average product of labor or APL = Q/L
This is the accounting measure of productivity.
- ⇒ **A numerical example:**

Units of labor (a)	Total product (b)	Marginal product (c)	Average product($d=b/a$)
0	0		
1	2000	2000	2000
2	3000	1000	1500
3	3500	500	1167
4	3800	300	950
5	3900	100	780

Marginal Product of Labor

- ⇒ The marginal product of labor (or MPL) refers to a company's increase in total production when one additional unit of labor is added (in most cases, one additional employee) and all other factors of production remain constant. In other words, the MPL is the additional output a company experiences after hiring another worker.
- ⇒ Marginal product curve is downward sloping

Types of Costs of Production

1. **Fixed Costs:** Expenses that remain constant regardless of output, e.g., rent, salaries, and equipment leases. Fixed costs are time-limited and become variable in the long term.
2. **Variable Costs:** Costs that change with production levels, e.g., raw materials, utilities, and direct labor. If production is zero, these costs are zero.
3. **Total Cost:** Sum of fixed and variable costs. For example, producing 1,000 shirts at \$9 each with a \$1,500 rent results in a total cost of \$10,500 (\$9,000 + \$1,500).
4. **Average Cost:** Total cost divided by the number of units produced. Companies aim to minimize average cost per unit to maximize profit.
5. **Marginal Cost:** The cost of producing one additional unit, primarily influenced by variable costs. It helps guide decisions on resource allocation for optimal profitability.
6. **Average Fixed Costs (*AFC*)**
7. **Average Variable Costs (*AVC*)**
8. **Average Total Costs (*ATC*)**
9. $ATC = AFC + AVC$

$$AFC = \frac{\text{Fixed cost}}{\text{Quantity}} = \frac{FC}{Q}$$

$$AVC = \frac{\text{Variable cost}}{\text{Quantity}} = \frac{VC}{Q}$$

$$ATC = \frac{\text{Total cost}}{\text{Quantity}} = \frac{TC}{Q}$$

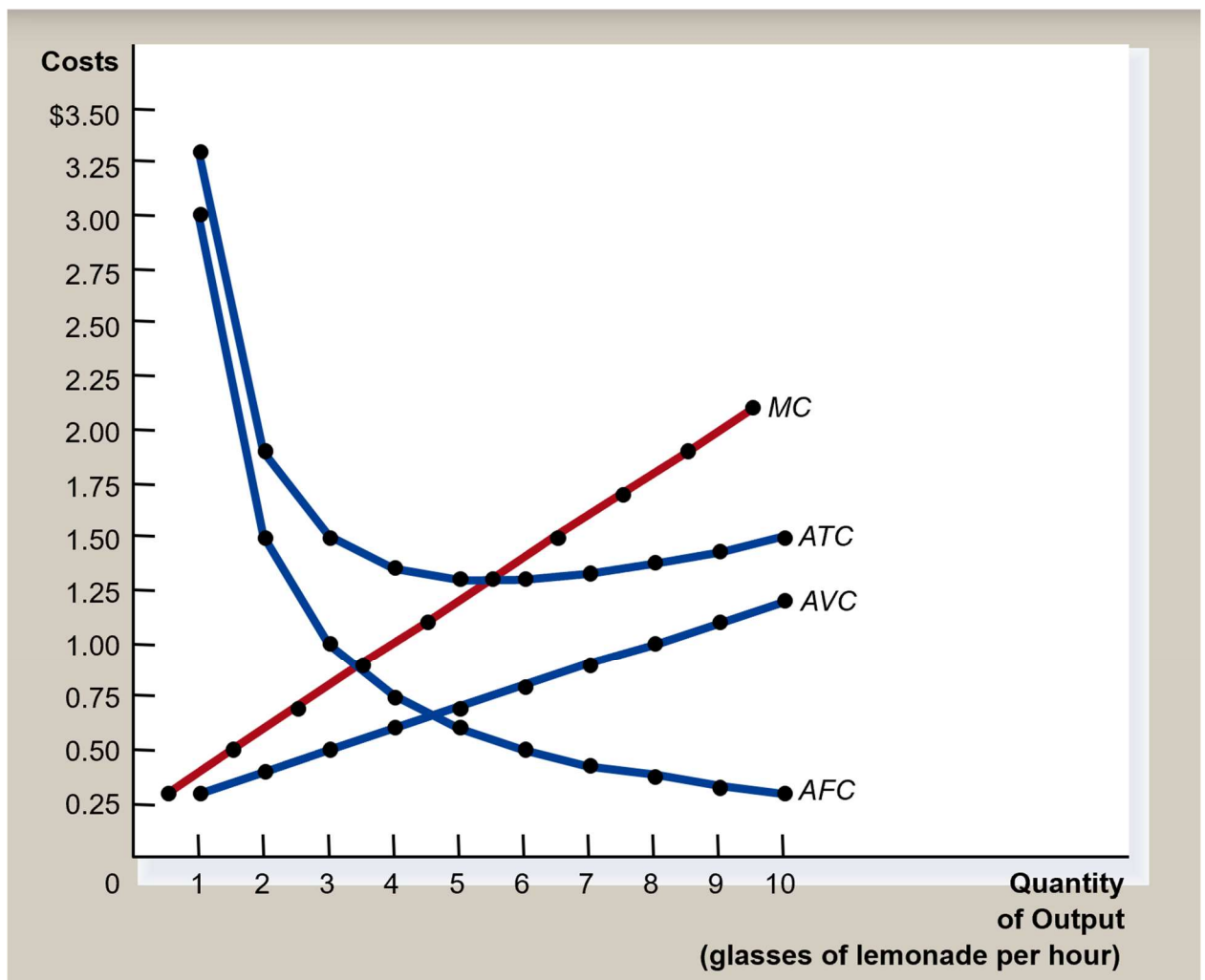
10. **Marginal Cost:** is the cost of producing one additional unit of output. It reflects the increase in total cost resulting from the production of that extra unit.

$$\text{Marginal Cost} = \text{Change in Total Expenses} \div \text{Change in Quantity of Units Produced}$$

$$MC = \frac{(\text{change in total cost})}{(\text{change in quantity})} = \frac{\Delta TC}{\Delta Q}$$

Quantity	Total Cost	Marginal Cost	Quantity	Total Cost	Marginal Cost
0	\$3.00	—	6	\$7.80	\$1.30
1	3.30	\$0.30	7	9.30	1.50
2	3.80	0.50	8	11.00	1.70
3	4.50	0.70	9	12.90	1.90
4	5.40	0.90	10	15.00	2.10
5	6.50	1.10			

Average-Cost, Marginal-Cost, Average Total Cost, Average Variable Costs Curves



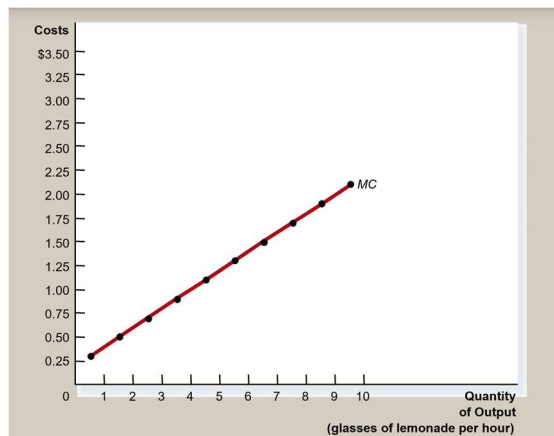
Cost Curves and Their Shapes

- **Marginal Cost (MC):** Rises with output due to diminishing marginal product.
- **Average Total Cost (ATC):** U-shaped curve:
 - **High at low output:** Fixed costs spread over few units.
 - **Declines as output increases,** then rises as average variable costs grow.
- **Efficient Scale:** Output level that minimizes ATC, located at the bottom of the U-shaped curve.

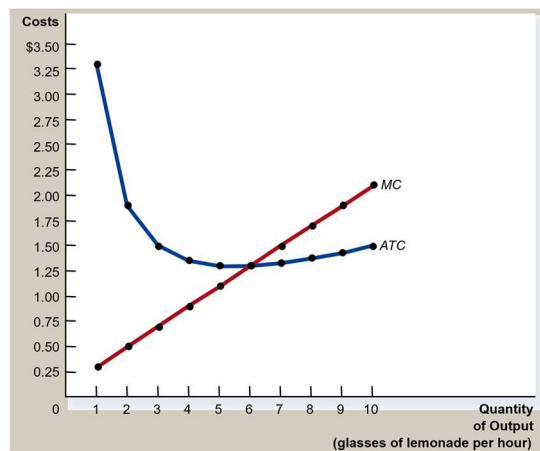
Relationship Between MC and ATC:

- **MC < ATC:** ATC is falling.
- **MC > ATC:** ATC is rising.
- **MC Curve Crosses ATC Curve:** At the efficient scale.

Marginal-Cost Curves



Average-Cost and Marginal-Cost Curves



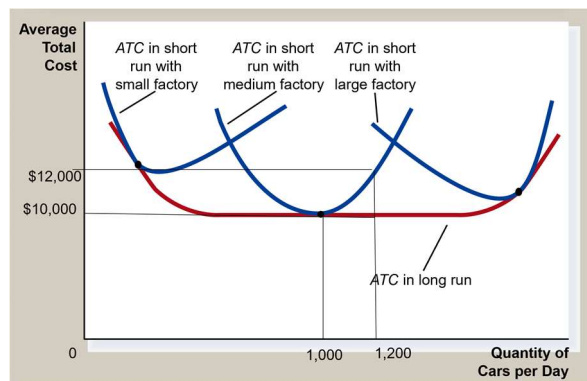
Typical Cost Curves

1. **Marginal cost rises** as output increases.
2. **Average total cost (ATC)** forms a **U-shape**.
3. **Marginal cost (MC)** intersects ATC at its minimum point.

COSTS IN THE SHORT RUN AND IN THE LONG RUN

1. Fixed vs. variable costs depend on the time horizon.
2. **Short run**: Some costs are fixed.
3. **Long run**: Fixed costs become variable, so long-run cost curves differ from short-run ones.

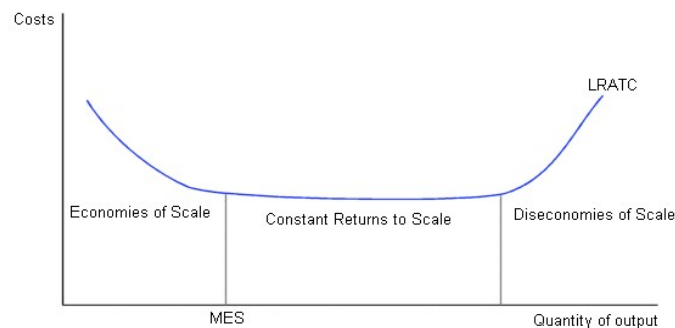
Average Total Cost in the Short and Long Run



Economies and diseconomies of scale

1. **Economies of scale**: Lower average cost as firm size grows (e.g., specialization, capital indivisibilities).
2. **Diseconomies of scale**: Higher average cost due to management and coordination challenges.
3. **Constant returns to scale**: Average cost remains unchanged with firm size.

Long-run average total cost (LRATC)



#Problem 1:

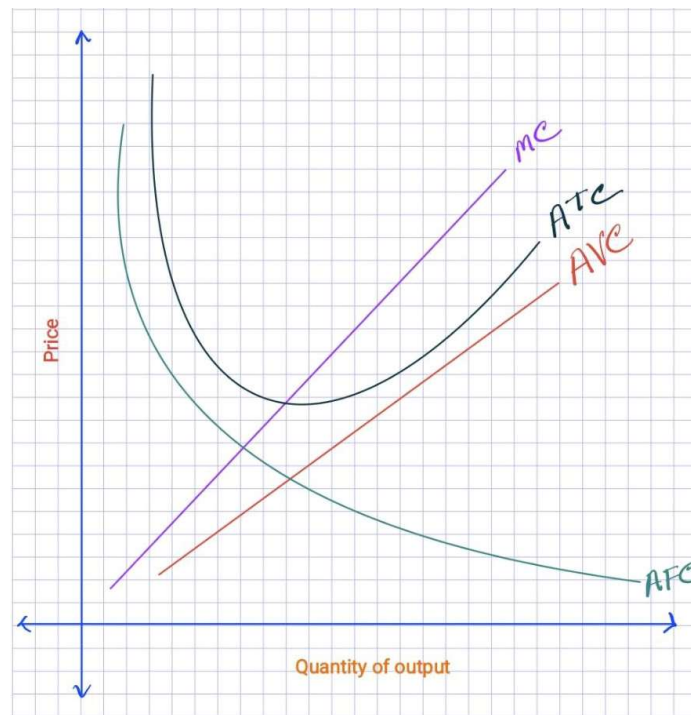
3. a) Give definitions for following: i) Fixed cost ii) Variable cost iii) Implicit cost iv) Explicit cost v) Accounting profit vi) Economic profit vii) marginal cost
- b) Draw MC, AVC, AFC, ATC in one diagram.
- c) Discuss the relationship between ATC and MC with the help of diagram.
- d) Discuss economies of scale, diseconomies of scale and constant return to scale with diagram.

(a)

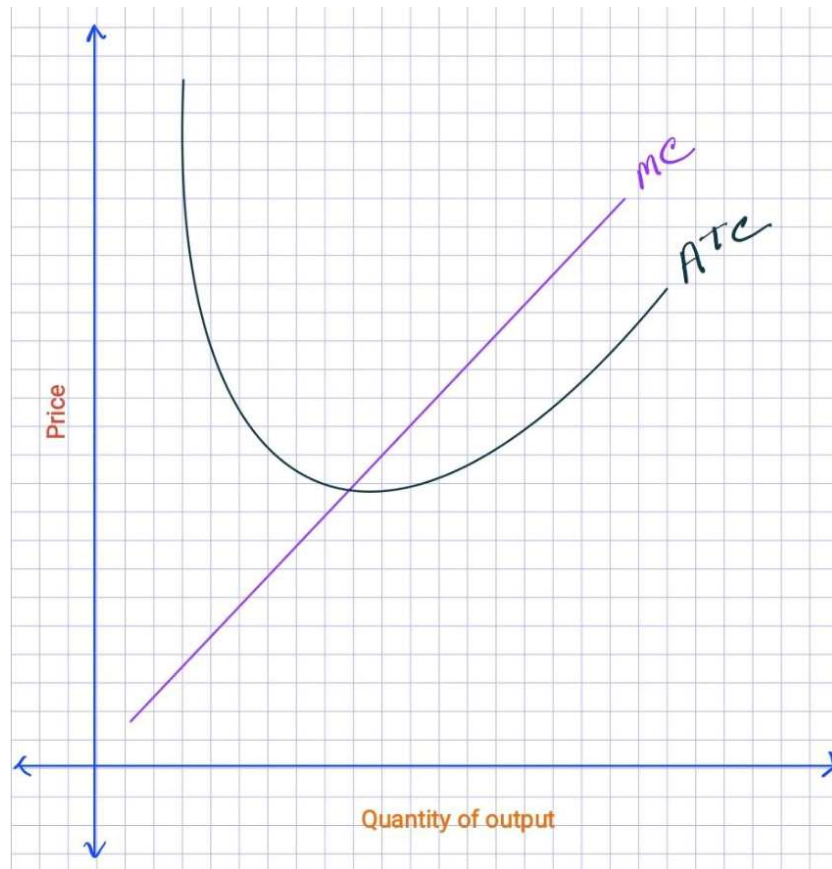
Definitions:

- i) **Fixed Cost (FC):** Costs that do not change with the level of output (e.g., rent, salaries).
- ii) **Variable Cost (VC):** Costs that vary with the level of output (e.g., raw materials, utilities).
- iii) **Implicit Cost:** Non-monetary opportunity costs of using resources owned by the firm (e.g., owner's time or foregone rent).
- iv) **Explicit Cost:** Direct, monetary payments made by a firm for inputs (e.g., wages, materials).
- v) **Accounting Profit:** Total revenue minus explicit costs.
- vi) **Economic Profit:** Total revenue minus explicit and implicit costs.
- vii) **Marginal Cost (MC):** The additional cost incurred by producing one more unit of output.

(b)



(c)



Relationship Between ATC and MC

- When $MC < ATC$, ATC is decreasing.
- When $MC > ATC$, ATC is increasing.
- **MC intersects ATC at its minimum point**, indicating where ATC stops falling and starts rising.

(d)

Economies of Scale, Diseconomies of Scale, and Constant Returns to Scale

1. **Economies of Scale:** As firm size increases, average cost decreases due to factors like specialization and efficient resource use.
2. **Diseconomies of Scale:** Beyond a certain size, average cost rises due to coordination and management challenges.
3. **Constant Returns to Scale:** Firm size changes, but average costs remain constant.

MCQ Practice

Topic-1: Production Possibility Frontier (PPF)

1. The Production Possibilities Frontier (PPF) of a nation shows -
 - a) how much people consume
 - b) how much production take place
 - c) the prices of the products
 - d) the population
2. Any point inside the PPF is called?
 - a. Unattainable point
 - b. Inefficient point
 - c. Efficient point
 - d. None of the above
3. Any point on the PPF is called?
 - a. Unattainable point
 - b. Inefficient point
 - c. Efficient point
 - d. None of the above

Topic-2: Demand, Supply and Equilibrium

1. Demand curve is—
 - a. Positively slopped
 - b. Negatively sloped
 - c. Upward sloping
 - d. Both (a) and (c)
2. Supply Curve is--
 - a. Positively slopped
 - b. Negatively sloped
 - c. Upward sloping
 - d. Both (a) and (c)
3. Ceteris paribus means:
 - a) Other thing being unequal
 - b) Other thing being equal
 - c) Other thing being inadequate
 - d) Both b & c
4. Which factor can lead to movement in the supply curve?
 - a) Product's Own Price
 - b) Input prices
 - c) Technology
 - d) Number of sellers

5. A decrease in demand means that:

- a) When price increases consumers are willing to purchase lesser quantities of the good
- b) Consumers are willing to purchase lesser quantities of the good at any given price
- c) When the price rises, consumers are willing to purchase greater quantities of the good
- d) Consumers cause the price drop by buying greater quantities of the good

6. Suppose that a scientific study just published demonstrates that having sugar makes people much healthier. How will this affect the equilibrium price and quantity of sugar in the market?

- a) The equilibrium price will increase and the equilibrium quantity will decrease
- b) The equilibrium price will decrease and the equilibrium quantity will increase
- c) Both the equilibrium quantity and price will increase
- d) Both the equilibrium quantity and price will decrease

7. In the market equilibrium diagram, price is in—

- a) Horizontal axis
- b) Vertical axis
- c) Both axis
- d) None of the above

8. Shortage in the market occurs at point when---

- a. quantity demand < quantity supply
- b. quantity demand > quantity supply
- c. quantity demand = quantity supply
- d. None of the above

9. Surplus in the market occurs at point when-

- a. quantity demand < quantity supply
- b. quantity demand > quantity supply
- c. quantity demand = quantity supply
- d. None of the above

10. If the price in a market happens to be above equilibrium price, there will be a _____ in the market, and the price will tend to _____.

- | | |
|-------------------|-------------------|
| a) surplus, fall | b) surplus, rise |
| c) shortage, fall | d) shortage, rise |

Topic-3: Elasticity

1. Perfectly elastic demand is _____
(a) horizontal (b) vertical (c) diagonal (d) downward
2. A perfectly inelastic demand is represented graphically by a:
a) relatively steep demand curve b) relatively flat demand curve
c) vertical demand curve d) horizontal demand curve
3. If there is a 1 percent price cut increases the quantity demanded by more than one percent then demand is _____
(a) elastic (b) inelastic (c) unit elastic (d) perfectly inelastic
4. Cross price elasticity of demand for complementary goods is _____
(a) positive (b) zero (c) negative (d) none
5. Suppose that two goods have a cross-price elasticity of demand of 0.8. This means that these goods are –
a) Normal b) Inferior
c) Complementary d) Substitute
6. Suppose that consumers' incomes rise by 2% and this causes demand for a good to increase by 6 %. What is the income elasticity of demand?
a) 2 b) 3
c) 0.3 d) 0.2
7. The more time consumers have to adjust to a price change, the _____ is the demand for that good.
(a) more inelastic (b) more elastic (c) perfectly elastic (d) none
8. The less time consumers have to adjust to a price change, the is the demand for that good.
a) more inelastic b) more perfectly elastic
c) more elastic d) none of the above
9. Demand tends to be more inelastic if the good is a _____
(a) luxury good (b) necessary good (c) both (d) none

Topic-4: Output and Costs

1. The cost of producing an extra unit of output is the.....
 - a) average variable cost
 - b) average fixed cost
 - c) average total cost
 - d) marginal cost
2. Efficient scale is the quantity that minimizes.....
 - a) average variable cost
 - b) average fixed cost
 - c) average total cost
 - d) marginal cost

PART B: Fill in the type of Cost that best completes each sentence from the below mentioned costs:

opportunity cost, total cost, fixed cost, variable cost, explicit costs, average total cost, average variable cost, accounting cost, and marginal cost

1. What you give up for taking some action is called theOpportunity cost
2. ...Fixed costs.... are those costs that do not vary with the quantity of output produced.
3. A cost that does not depend on the quantity produced is a(n)fixed cost.....
4. The cost of producing an extra unit of output is the.....marginal cost.....
5. Theaverage total cost..... curve is U-shaped.
6. Themarginal cost..... crosses the average-total-cost curve at the efficient scale.
7.average total cost..... is falling when marginal cost is below it and rising when marginal cost is above it.
8. In the ice-cream industry in the short run,variable cost..... includes the cost of cream and sugar but not the cost of the factory.

True/False Practice

- True 1. Microeconomics deals with the behavior of individual economic units.
- False 2. Changes in a product's own price represents **shift** (**movement**) in the demand curve.
- False 3. Market demand is the sum of all individual **supplies** (**demands**) at each possible price.
- False 4. Whenever marginal cost is less than average total cost, average total cost is **rising** (**falling**).
- False 5. Demand tends to be more **inelastic** (**elastic**), if the good is a luxury good.
- False 6. Demand tends to be more **elastic** (**inelastic**), if the good is a necessary good.
- False 7. The short-run is a time frame in which the quantities of all inputs can be **varied** (**fixed**).
- False 8. The Long-run is a time frame in which the quantities of all inputs can be **fixed** (**Varied**).
- False 9. Suppose that consumers' income rise causes demand for a good to decrease, the good is a **normal** (**inferior**) good.

THE END